ELECTRICAL LEGEND

(NOTE: NOT ALL SYMBOLS SHOWN ARE USED ON THESE DRAWINGS)

- ONE LINE SYMBOLS -

SYMBOL	DESCRIPTION
	CIRCUIT BREAKER
	DRAW-OUT CIRCUIT BREAKER (MOLDED INSULATED CASE)
52	DRAW-OUT POWER CIRCUIT BREAKER
	CONTROL FUSE
	FUSE WITH SWITCH
	SWITCH
PANEL XXX	PANELBOARD
ATS-XX XXX A	AUTOMATIC TRANSFER SWITCH
100A4G	FEEDER DESIGNATION, SEE FEEDER SCHEDULE
	AUTOMATIC TRANSFER SWITCH WITH BY-PASS
E	ENGINE GENERATOR
TX-XXXX	TRANSFORMER
	ENCLOSED BUSWAY
▭	GROUND BUS
	WEATHERHEAD
Ⓧ	MOTOR
△	DELTA CONNECTION
Y	WYE CONNECTION
	GROUNDED WYE CONNECTION
	GROUNDED WYE CONNECTION WITH RESISTOR GROUND
	GROUNDED WYE CONNECTION WITH REACTOR GROUND
Ⓜ	METERING DEVICE
	CURRENT TRANSFORMER
	POTENTIAL TRANSFORMER
	LOAD-BREAK CONNECTOR
##	PROTECTIVE RELAY DEVICE
Ⓚ	KEY INTERLOCK
2400V, 100A 10SEC. RESISTOR	RESISTOR
	CONTACT NORMALLY OPEN
	CONTACT NORMALLY CLOSED
	CAPACITOR
	SINGLE BATTERY
	MULTIPLE BATTERIES
	LIGHTNING ARRESTOR
	THERMAL ELEMENT, OVERLOAD RELAY

- GENERAL -

SYMBOL	DESCRIPTION
	BRANCH CIRCUIT HOME RUN TO PANELBOARD, DESIGNATION INDICATES PANEL AND CIRCUIT NUMBERS
~	CONTROL WIRING
—	LIGHTING, ONE-LINE, AND POWER CIRCUITING
– –	LIGHTING, ONE-LINE, AND POWER CIRCUITING (UNDERGROUND)
~	FLEXIBLE CONDUIT
	CONDUIT BREAK SYMBOL
	CONDUIT CAP
	CONDUIT CHANGE IN ELEVATION
—●	CONDUIT STUB DOWN (OUT OF DRAWING LIMITS)
—○	CONDUIT STUB UP (OUT OF DRAWING LIMITS)
Ⓙ	JUNCTION BOX
Ⓙ	WALL MOUNTED JUNCTION BOX
Ⓙ	FLOOR MOUNTED JUNCTION BOX
	PUSH BUTTON A = ABORT DA = DURESS ALARM EPO = EMERGENCY POWER OFF IC = INTERCOM ST = SHUNT TRIP
$S^{X}_{\#}$	SWITCH SYMBOL (#) SINGLE POLE (IF BLANK) 2 = DOUBLE POLE 3 = THREE-WAY 4 = FOUR-WAY AS = ADJUSTABLE SPEED D = DIMMER K = KEY OPERATED LV = LOW VOLTAGE M = MANUAL MOTOR SWITCH OS = OCCUPANCY SENSOR P = WITH PILOT LIGHT T = TIMER TO = THERMAL OVERLOAD VS = VACANCY SENSOR WP = WEATHERPROOF x = SMALL LETTER - LUMINAIRES CONTROLLED XP = EXPLOSION PROOF
S S	DUAL SWITCH
$	INTERCOM SWITCH

- FIRE ALARM DEVICES -

SYMBOL	DESCRIPTION
B	BELL
D D	DUCT SMOKE DETECTOR
	FIRE FIGHTER'S TELEPHONE JACK
H	COMBINATION FIRE HORN/STROBE LIGHT
S	COMBINATION FIRE SPEAKER/STROBE LIGHT
	FIRE ALARM STROBE LIGHT
	FIRE ALARM STROBE, CEILING MOUNT
Ⓢ	FIRE ALARM STROBE/SPEAKER, CEILING MOUNT
H	FIRE HORN
H	DUAL PROJECTION FIRE HORN
F	MANUAL PULL STATION
■	MAGNETIC DOOR HOLD OPEN
D F X	DETECTOR / DETECTOR UNDER FLOOR F = FLAME I = IONIZATION TYPE P = PHOTOELECTRIC TYPE T = THERMAL TYPE
Ⓑ	BEAM DETECTOR
	PULL STATION/TELEPHONE JACK
Ⓢ	DELUGE VALVE

- SPECIAL SYSTEMS DEVICES -

SYMBOL	DESCRIPTION
▽	DATA OUTLET
	COMBINATION TELEPHONE/DATA OUTLET
▼	TELEPHONE OUTLET
	TELEVISION JACK
	CEILING MOUNTED DATA OUTLET
	CEILING MOUNTED TELEPHONE/DATA OUTLET
	CEILING MOUNTED TELEPHONE OUTLET
	FLOOR MOUNTED DATA OUTLET
	FLOOR MOUNTED TELEPHONE/DATA OUTLET
	FLOOR MOUNTED TELEPHONE OUTLET
TTB	TELEPHONE TERMINAL BOARD
DTB	DATA TERMINAL BOARD
M	MICROPHONE OUTLET
Ⓢ X	SPEAKER V = WITH INTEGRAL VOLUME CONTROL
S	WALL MOUNTED SPEAKER
V	WALL MOUNTED VOLUME CONTROL
C	CLOCK RECEPTACLE OUTLET
	SECURITY CAMERA
Ⓣ	THERMOSTAT
XX	SPECIAL SYSTEMS SYMBOLS: AB = ABORT BUTTON AMP = AMPLIFIER ANN = ANNUNCIATOR ASD = ADJUSTABLE SPEED DRIVE ATS = AUTOMATIC TRANSFER SWITCH B = BUZZER BMS = BALANCED MAGNETIC STRIP CAM = CONTROL ADDRESSABLE MODULE CH = CHIME CK = CARD READER/KEYPAD CR = CARD READER DA = DURESS ALARM PUSH-BUTTON DC = DOOR CONTACT DE = DELAYED EGRESS DMA = DOOR MANAGEMENT ALARM DO = DOOR OPERATOR DS = DOOR STRIKE DTC = DATA TERMINAL CABINET DVR = DIGITAL VIDEO RECORDER EC = ELECTRICAL CONTACTOR EL = ELECTRIC LOCK ES = ELECTRIC STRIKE F = MANUAL PULL STATION FAA = FIRE ALARM ANNUNCIATOR FACP = FIRE ALARM CONTROL PANEL FS = FLOW SWITCH FSD = FIRE SMOKE DAMPER GB = GLASSBREAK DETECTOR HC = HANDICAP PUSH-BUTTON IC = INTERCOM CALL STATION IP = INDICATOR PANEL KP = KEYPAD MAM = MONITOR ADDRESSABLE MODULE MH = MANHOLE MD = MOTORIZED DAMPER MS = MOTION SENSOR ML = MAGNETIC LOCK NCMS = NURSE CALL MASTER STATION CM-CENTRALIZED PHYSIOLOGICAL MONITORING STATION NCT = NURSE CALL TERMINAL CABINET OS = OCCUPANCY SENSOR PA = PUBLIC ADDRESS PB = PULLBOX PIR = PASSIVE INFRARED DETECTOR PM = POWER QUALITY METER PS = PRESSURE SWITCH R = REMOTE INDICATING LIGHT RT = REMOTE TEST STATION RX = REQUEST TO EXIT PUSH-BUTTON S = WALL MOUNTED SPEAKER SAP = SECURITY ALARM PANEL SPD = SURGE PROTECTIVE DEVICE TC = TIME CLOCK TS = TAMPER SWITCH VFD = VARIABLE FREQUENCY DRIVE VM = VIDEO MONITOR VS = VACANCY SENSOR

- POWER -

SYMBOL	DESCRIPTION
⏻	DUPLEX RECEPTACLE D = DEDICATED CIRCUIT IG = ISOLATED GROUND DEVICE ⏻ = AUTOMATICALLY CONTROLLED RECEPTACLE
GFI	GFI = GROUND FAULT CIRCUIT INTERRUPTER
	FLOOR MOUNTED DUPLEX RECEPTACLE
	FLOOR MOUNTED FOURPLEX RECEPTACLE
	FLOOR MOUNTED SPECIAL PURPOSE RECEPTACLE
	CEILING MOUNTED DUPLEX RECEPTACLE
	CEILING MOUNTED FOURPLEX RECEPTACLE
	CEILING MOUNTED SPECIAL PURPOSE RECEPTACLE
	FOURPLEX RECEPTACLE
	RANGE RECEPTACLE
	SINGLE RECEPTACLE
	SPECIAL PURPOSE RECEPTACLE
	SWITCHED RECEPTACLE
	DUPLEX EMERGENCY/CRITICAL
	FOURPLEX EMERGENCY/CRITICAL
	DUPLEX-2PORT USB
	4PORT USB
	DISCONNECT SWITCH
	FUSED DISCONNECT SWITCH
	ENCLOSED CIRCUIT BREAKER
⊠	MOTOR STARTER
⊠	COMBINATION MOTOR STARTER
S	COMBINATION SWITCH AND RECEPTACLE
⧓	CONDUIT SEAL OFF
⊙	FIRE RATED POKE-THROUGH
	PARTITION CIRCUIT SPLIT
◪	POWER POLE
W	SURFACE RACEWAY
M	METER

- NURSE CALL -

SYMBOL	DESCRIPTION
CB	CODE BLUE STATION
CP	CODE PINK STATION
EP	EMERGENCY CALL PULL STATION
SP	SINGLE PATIENT STATION
SA	STAFF ASSIST STATION
AI	AUXILIARY INPUT STATION
DS	DUTY STATION
SS	STAFF STATION
BI	BED INTERFACE STATION
2B	TWO BUTTON STATION
BM	BED MANAGEMENT STATION
D	DOME LIGHT
Z	ZONE LIGHT INTERSECTION
RL	REMOTE LOCATOR
PM	PATIENT MONITOR

- LIGHTING -

(REFER TO LUMINAIRE SCHEDULE)

SYMBOL	DESCRIPTION
X * #	LUMINAIRES X = FIXTURE DESIGNATION # = BRANCH CIRCUIT NUMBER * = SWITCH LEG IDENTIFIER
●	SHADING INDICATES LUMINAIRE ON LIFE SAFETY
	SHADING INDICATES PORTION OF LUMINAIRE ON LIFE SAFETY
	WALL MOUNTED LUMINAIRE
	STRIP LIGHT
	STRIP LIGHT WITH LIFE SAFETY
	POLE MOUNTED LUMINAIRE (QUANTITY OF LUMINAIRES PER POLE AS INDICATED ON PLANS)
○ □	DOWNLIGHT LUMINAIRE
◑	WALL WASHER LUMINAIRE
	ADJUSTABLE LUMINAIRE
	PENDANT LUMINAIRE
	TRACK LIGHTING
Ⓟ	PHOTOCELL
	EXIT LIGHT (WITH FACES AND DIRECTION ARROWS INDICATED)
	WALL MOUNTED EXIT LIGHT (WITH FACES AND DIRECTION ARROWS INDICATED)
	WALL MOUNTED BATTERY PACK EMERGENCY LIGHT

- GROUNDING SYMBOLS -

SYMBOL	DESCRIPTION
⊗	GROUND TEST WELL
—	GROUNDING CONDUCTOR
⦿	LIGHTNING PROTECTION AIR TERMINAL
—●	BONDING POINT
G	GROUND BAR
⏚	ELECTRICAL GROUND
	GROUND ROD
	GROUND ROD WITH INSPECTION TEST WELL
	PIGTAIL

- EQUIPMENT -

SYMBOL	DESCRIPTION
	DISTRIBUTION PANEL
	EXISTING DISTRIBUTION PANEL
	NEW PANEL, FLUSH MOUNTED
	EXISTING PANEL, FLUSH MOUNTED
	NEW PANEL, SURFACE MOUNTED
	EXISTING PANEL, SURFACE
T	TRANSFORMER

APPLICABLE CODES

AHJ: STATE OF COLORADO

FIRE AUTHORITY: DENVER FIRE

REMODEL ☒ NEW ☐

YEAR	CODE
2024	INTERNATIONAL BUILDING CODE
2024	INTERNATIONAL MECHANICAL CODE
~~2024~~	~~INTERNATIONAL PLUMBING CODE~~
2024	INTERNATIONAL ENERGY CONSERVATION CODE
~~2024~~	~~INTERNATIONAL FIRE CODE~~
2023	NATIONAL ELECTRICAL CODE
2025	DENVER BUILDING AND FIRE CODES

IS THE BUILDING FULLY SPRINKLERED? ☐ Yes ☒ No

- GRAPHIC SYMBOLS -

SYMBOL	DESCRIPTION
1	KEY NOTE
#	REVISION NUMBER
X	DETAIL NOTE
X	X = DENOTES ALL LUMINAIRES IN THE RESPECTIVE AREA ARE THE TYPE INDICATED, REFER TO LUMINAIRE SCHEDULE
TAG	PANEL FLAG
TAG	OWNER EQUIPMENT TAG
S# X	LIGHTING CONTROL SEQUENCE INDICATION, SEE LIGHTING CONTROL SEQUENCE OF OPERATION SCHEDULE FOR INFORMATION X = DENOTES ILLUMINATION SET POINT
TAG	MECHANICAL EQUIPMENT TAG
▭	SHADING INDICATES EQUIPMENT
▨	HATCHING INDICATES ITEM(S) TO BE REMOVED
A100	ROOM NUMBER
N	NORTH ARROW
1 E2.1	DETAIL BUBBLE DETAIL NUMBER SHEET NUMBER - WHERE DETAIL IS SHOWN
2 E2.2	SECTION CUT SECTION NUMBER/LETTER SHEET NUMBER - WHERE SECTION IS SHOWN

- ABBREVIATIONS -

(E)	EXISTING
(N)	NEW
(PART)	PARTIAL CIRCUIT
A	AMP
AC	ABOVE COUNTER TOP
AF	AMP FRAME, AMP FUSE
AFF	ABOVE FINISHED FLOOR
AFG	ABOVE FINISHED GRADE
AHJ	AUTHORITY HAVING JURISDICTION
AIC	AMPS INTERRUPTING CAPACITY
AL	ALUMINUM
APL	APPLIANCE
AS	AMP SWITCH
AT	AMP TRIP
ATS	AUTOMATIC TRANSFER SWITCH
AV	AUDIOVISUAL
BJ	BONDING JUMPER
C	CONDUIT
CB	CIRCUIT BREAKER
CCT	CIRCUIT
CCTV	CLOSED CIRCUIT TV
CLG	CEILING
CU	COPPER
DISC	DISCONNECT
DIST	DISTRIBUTION
E/G	ENGINE/GENERATOR
ELEC	ELECTRIC, ELECTRICAL
ELEV	ELEVATOR ELEVATION
EM	EMERGENCY
EQPT	EQUIPMENT
F	FUSE
FA	FIRE ALARM
FAA	FIRE ALARM ANNUNCIATOR
FACP	FIRE ALARM CONTROL PANEL
FIXT	FIXTURE
FO	FIBER OPTIC
FLR	FLOOR
FLUOR	FLUORESCENT
G	GROUND (EQUIPMENT)
GEN	GENERATOR
GFI	GROUND FAULT CIRCUIT INTERRUPTER
GF	GROUND FAULT PROTECTION
GND	GROUND
HC	HANDICAPPED
HG	HOSPITAL GRADE
HP	HORSEPOWER
HV	HIGH VOLTAGE
INC	INCANDESCENT
IDF	INTERMEDIATE DISTRIBUTION FRAME
IG	ISOLATED GROUND
JB	JUNCTION BOX
LED	LIGHT EMITTING DIODE
LTG	LIGHTING
LV	LOW VOLTAGE
MATV	MASTER ANTENNA T.V.
MCB	MAIN CIRCUIT BREAKER
MCC	MOTOR CONTROL CENTER
MDF	MAIN DISTRIBUTION FRAME
MH	MANHOLE
MLO	MAIN LUGS ONLY
MOP	METHOD OF PROCEDURE
MTD	MOUNTED
MTG	MOUNTING
MTR	MOTOR
MV	MEDIUM VOLTAGE
N	NEUTRAL
NC	NORMALLY CLOSED
NEC	NATIONAL ELECTRICAL CODE
NF	NON-FUSED
NIC	NOT IN CONTRACT
NL	NIGHT LIGHT
NO	NORMALLY OPEN
OHE	OVERHEAD ELECTRIC
P	POLE
PA	PUBLIC ADDRESS
PB	PULL BOX
PH	PHASE
PNL	PANEL
PWR	POWER
RCPT	RECEPTACLE
SBC	STRANDED BARE COPPER
SPD	SURGE PROTECTIVE DEVICE
SW	SWITCH
TEL	TELEPHONE
TP	TAMPERPROOF
TV	TELEVISION
UF	UNDERFLOOR
UNO	UNLESS NOTED OTHERWISE
UPS	UNINTERRUPTIBLE POWER SUPPLY
V	VOLTAGE
VFD	VARIABLE FREQUENCY DRIVE
VP	VAPOR PROOF
W	WIRE
WAP	WIRELESS ACCESS POINT
WG	WIRE GUARD
WP	WEATHERPROOF
XFMR	TRANSFORMER
XP	EXPLOSION PROOF

REV: 10/21/2024

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GENERAL PROJECT NOTES

1. LIGHT LINE WEIGHT INDICATES EXISTING. HEAVY LINE WEIGHT INDICATES NEW CONSTRUCTION.
2. SPECIFICATIONS ARE A PART OF THE CONSTRUCTION DOCUMENTS. SHOULD ANY CONFLICT ARISE BETWEEN THE DRAWINGS AND SPECIFICATIONS, BRING SUCH CONFLICT TO THE ATTENTION OF THE ENGINEER FOR RESOLUTION. UNLESS OTHERWISE DIRECTED BY ENGINEER, THE MOST STRINGENT REQUIREMENT WILL PREVAIL.
3. DATA ON THE DRAWINGS IS AS EXACT AS COULD BE REASONABLY SECURED. ABSOLUTE ACCURACY IS NOT GUARANTEED. VERIFY EXACT LOCATIONS, MEASUREMENTS, LEVELS, SPACE REQUIREMENTS, POTENTIAL CONFLICTS WITH OTHER TRADES. ADAPT WORK TO ACTUAL CONDITIONS AT THE SITE. VISIT THE SITE TO BECOME THOROUGHLY FAMILIAR WITH THE ACTUAL CONDITIONS OF THIS PROJECT BEFORE SUBMITTING COSTS. THESE DRAWINGS ARE DIAGRAMMATIC IN NATURE; DO NOT SCALE. THESE DRAWINGS DO NOT SHOW MATERIALS FOR A COMPLETE INSTALLATION; PLAN THE INSTALLATION AND LAYOUT OF THE WORK AS DIAGRAMMED IN THESE DOCUMENTS. REFER TO FLOOR PLANS, SCHEMATICS AND DIAGRAMS OF OTHER TRADES FOR ELECTRICAL REQUIREMENTS, BRANCH CIRCUITS AND OTHER ELECTRICAL CONNECTIONS NOT INDICATED ON THESE DOCUMENTS.
4. FIRE-SEAL ALL PENETRATIONS THROUGH RATED ASSEMBLIES WITH MATERIALS CAPABLE OF PREVENTING THE PASSAGE OF FLAMES AND HOT GASES WHEN SUBJECTED TO THE REQUIREMENTS OF THE TEST STANDARD SPECIFIC FOR FIRE STOPS ASTM E814.
5. JUNCTION BOXES FOR LUMINAIRES AND OUTLETS ARE NOT INDICATED. PROVIDE THE PROPER QUANTITY OF JUNCTION BOXES TO MEET LOCAL CODE AND NATIONAL ELECTRICAL CODE.
6. EXECUTE THE WORK IN ACCORDANCE WITH SUPPORTING OBJECTS FOR SEISMIC ZONE REQUIRED BY STATE AND LOCAL CODES FOR ALL CEILING ATTACHED OBJECTS AND FLOOR ATTACHED EQUIPMENT INCLUDING, BUT NOT LIMITED TO: PENDANT LIGHTING FIXTURES, GENERAL LIGHTING, MULTIPLE RACEWAYS, GENERATOR, TRANSFORMER, ELECTRICAL SWITCHGEAR, SWITCHBOARDS AND OTHER ELECTRICAL EQUIPMENT.
7. COORDINATE AND SCHEDULE ALL ELECTRICAL OUTAGES AND OUTAGE TIME FRAMES WITH OWNER'S REPRESENTATIVE AT LEAST TWO WEEKS PRIOR TO OUTAGE. ACCOMPLISH ALL PREP WORK PRIOR TO OUTAGE TO MINIMIZE THE OUTAGE TIME FRAME. SUBMIT A DETAILED METHOD OF PROCEDURE (MOP) FOR ALL OUTAGES FOUR WEEKS PRIOR TO OUTAGE. PERFORM OUTAGES DURING OFF HOURS AT A TIME ACCEPTABLE BY THE OWNER.
8. UPDATE PANEL SCHEDULES TO REFLECT INSTALLED CONDITIONS AFTER CONSTRUCTION IS COMPLETE.
9. UNLESS OTHERWISE INDICATED, BRANCH CIRCUITS SHALL BE 3/4"C (2#12, #12G) PROTECTED BY 20-AMPERE CIRCUIT BREAKERS. INCREASE CONDUCTOR SIZE TO ACCOUNT FOR VOLTAGE DROP FOR ALL 120-VOLT CIRCUITS OVER 75 FEET, AND ALL 277-VOLT CIRCUITS OVER 150 FEET TO THE FIRST OUTLET. CONDUCTOR SIZE SHALL BE UNIFORM FOR THE ENTIRE LENGTH OF THE CIRCUIT UNLESS NOTED OTHERWISE. HOMERUNS WHICH INDICATE UPGRADING CIRCUIT CONDUCTORS FOR VOLTAGE DROP, E.G., #10AWG WIRE ON 20-AMPERE CIRCUIT, SHALL HAVE THE CONDUCTOR SIZE INDICATED CARRIED THROUGHOUT THE CIRCUIT TO ALL JUNCTION BOXES UP TO AND INCLUDING THE J-BOX NEAREST THE LAST DEVICE OR LUMINAIRE.
10. THESE DRAWINGS HAVE BEEN PREPARED BASED ON INFORMATION PROVIDED BY OTHERS. DATA PRESENTED ON THIS DRAWING IS AS ACCURATE AS CAN BE DETERMINED, BUT ACCURACY IS NOT GUARANTEED. WHILE THIS INFORMATION IS BELIEVED TO BE RELIABLE, THE ENGINEER IS NEITHER RESPONSIBLE FOR ITS ACCURACY, NOR ERRORS NOR OMISSIONS WHICH MAY HAVE BEEN INCORPORATED INTO THESE DOCUMENTS. FIELD VERIFICATION OF ALL AFFECTED COMPONENTS IS REQUIRED.
11. PATCH AND PAINT ANY DAMAGED SURFACES DUE TO DEMOLITION AND CONSTRUCTION TO MATCH EXISTING CONDITIONS. SEAL ALL PENETRATIONS THROUGH RATED CEILINGS AND WALLS WITH UL LISTED SEALANTS AND FIRESTOPPING MATERIALS.
12. UP-TO-DATE ELECTRICAL RECORD DRAWINGS ARE NOT AVAILABLE FOR THIS PROJECT. INFORMATION FOR EXISTING CIRCUITRY IS BASED ON EXISTING PANEL DIRECTORIES, AVAILABLE DRAWINGS, AND ASSUMPTIONS. LOCATIONS AND INFORMATION FOR EXISTING ELECTRICAL DEVICES AND EQUIPMENT SHOWN ON THESE DOCUMENTS ARE APPROXIMATE AND WERE DERIVED FROM FIELD OBSERVATION AND AVAILABLE RECORD DRAWINGS. VERIFY ACTUAL FIELD CONDITIONS PRIOR TO STARTING WORK.

DEMOLITION PROJECT NOTES

1. WORK SHOWN HATCHED IS EXISTING TO BE REMOVED. REMOVE ALL DEVICES ON WALLS WHICH ARE REMOVED. WHERE DEVICES ARE REMOVED, REMOVE ASSOCIATED CIRCUIT CONDUCTORS AND CONDUIT BACK TO THE SERVING PANELBOARD OR NEXT DEVICE ON THE CIRCUIT WHICH REMAINS (MAINTAIN CIRCUIT CONTINUITY).
2. UNLESS OTHERWISE INDICATED, DEMOLITION WASTE BECOMES PROPERTY OF CONTRACTOR. REMOVE DEMOLITION WASTE MATERIALS FROM PROJECT SITE AND DISPOSE OF THEM IN AN APPROVED CONSTRUCTION AND DEMOLITION WASTE LANDFILL ACCEPTABLE TO AUTHORITIES HAVING JURISDICTION. COORDINATE DISPOSAL LOCATION WITH OWNERS REPRESENTATIVE PRIOR TO DISPOSING OF ANY REMOVED ELECTRICAL EQUIPMENT.
3. EXISTING CIRCUIT INFORMATION SHOWN ON THE PLANS IS FOR REFERENCE ONLY. TRACE ALL EXISTING CIRCUITS AFFECTED BY THIS WORK TO CONFIRM SOURCE PANEL AND CIRCUIT NUMBERS.
4. INFORMATION ON THE DRAWINGS HAS BEEN TAKEN FROM RECORD DRAWINGS. DEMOLITION DRAWINGS ARE INTENDED TO GIVE A COMMON BASIS FOR WORK. VISIT THE SITE, PRIOR TO FINAL PRICING AND BECOME FAMILIAR WITH THE EXTENT OF WORK REQUIRED. NO EXTRAS WILL BE ALLOWED FOR ALTERATIONS OF A FORESEEABLE NATURE REQUIRED TO ACHIEVE THE END RESULT AS INDICATED BY THE CONTRACT DOCUMENTS.
5. MAINTAIN FIRE ALARM SYSTEM OPERATION IN ALL AREAS OUTSIDE OF THE AREA OF WORK DURING CONSTRUCTION.

LIGHTING PROJECT NOTES

1. EXISTING CIRCUIT INFORMATION SHOWN ON THE PLANS IS FOR REFERENCE ONLY. TRACE ALL EXISTING CIRCUITS AFFECTED BY THIS WORK TO CONFIRM SOURCE PANEL AND CIRCUIT NUMBERS.
2. UPPERCASE LETTER DENOTES LUMINAIRE TYPE. LOWERCASE LETTER INDICATES CONTROL ZONE.
3. EMERGENCY LUMINAIRES SHALL BE SWITCHED WITH THE REST OF THE LUMINAIRES IN THE ROOM OR AS INDICATED ON THE DRAWINGS BY LIGHTING CONTROL ZONE. PROVIDE ONE EMERGENCY TRANSFER RELAY PER CONTROL ZONE. RELAY SHALL BE INSTALLED IN AN ACCESSIBLE LOCATION.
4. TO ACHIEVE THE INDICATED SEQUENCE OF OPERATION, PROVIDE ALL COMPONENTS AND DIMMING CABLING REQUIRED FOR THE LIGHTING CONTROL SYSTEM, WHETHER INDICATED OR NOT. REFER TO LIGHTING CONTROL SEQUENCE OF OPERATION SCHEDULE FOR ADDITIONAL INFORMATION.
5. PROVIDE DIMMERS COMPATIBLE WITH LED DRIVERS AND POWER SUPPLIES PER MANUFACTURER'S RECOMMENDATIONS. PROVIDE COMPONENTS AND ACCESSORIES TO ACCOMPLISH COMPLETE DIMMING FUNCTIONALITY AS SHOWN ON PLANS AND NOTED IN THE LUMINAIRE SCHEDULE.
6. FOR LIGHT DIMMING BRANCH CIRCUITS, PROVIDE A DEDICATED NEUTRAL CONDUCTOR FOR EACH BRANCH CIRCUIT CURRENT CARRYING CONDUCTOR. CIRCUITS MAY BE COMBINED INTO HOMERUNS OF UP TO SIX CURRENT CARRYING CONDUCTORS WITH PROPER DE-RATING AS REQUIRED BY NEC. PROVIDE FULL LENGTH STRIPING FOR DEDICATED NEUTRALS TO MATCH ASSOCIATED CURRENT CARRYING CONDUCTOR COLOR.
7. LIGHTING CONTROLS COMMISSIONING SHALL COMPLY WITH IECC 2024, INCLUDING BUT NOT LIMITED TO COMMISSIONING DOCUMENTS, SYSTEM FUNCTIONAL TESTING, AND RECORD DOCUMENTS. REFER TO SPECIFICATIONS FOR ADDITIONAL INFORMATION.

REV	DATE	DESCRIPTION
1	09/21/26	RESPONSE TO COMMENTS

SHARA RENEE JOHNSON
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09.21.2026

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Mechanical • Electrical
Industrial • Sustainability

METRO STATE
UNIVERSITY

CAVEA LIGHTING UPGRADE
ELECTRICAL LEGEND

Date:	JULY 17, 2026
Scale:	None
Designed By:	SRJ
Drawn By:	MLB
Approved By:	SKB
Project No:	21607
Project Status:	100% CD

RMH-21607

SHT NO: E-001

REVISION 1

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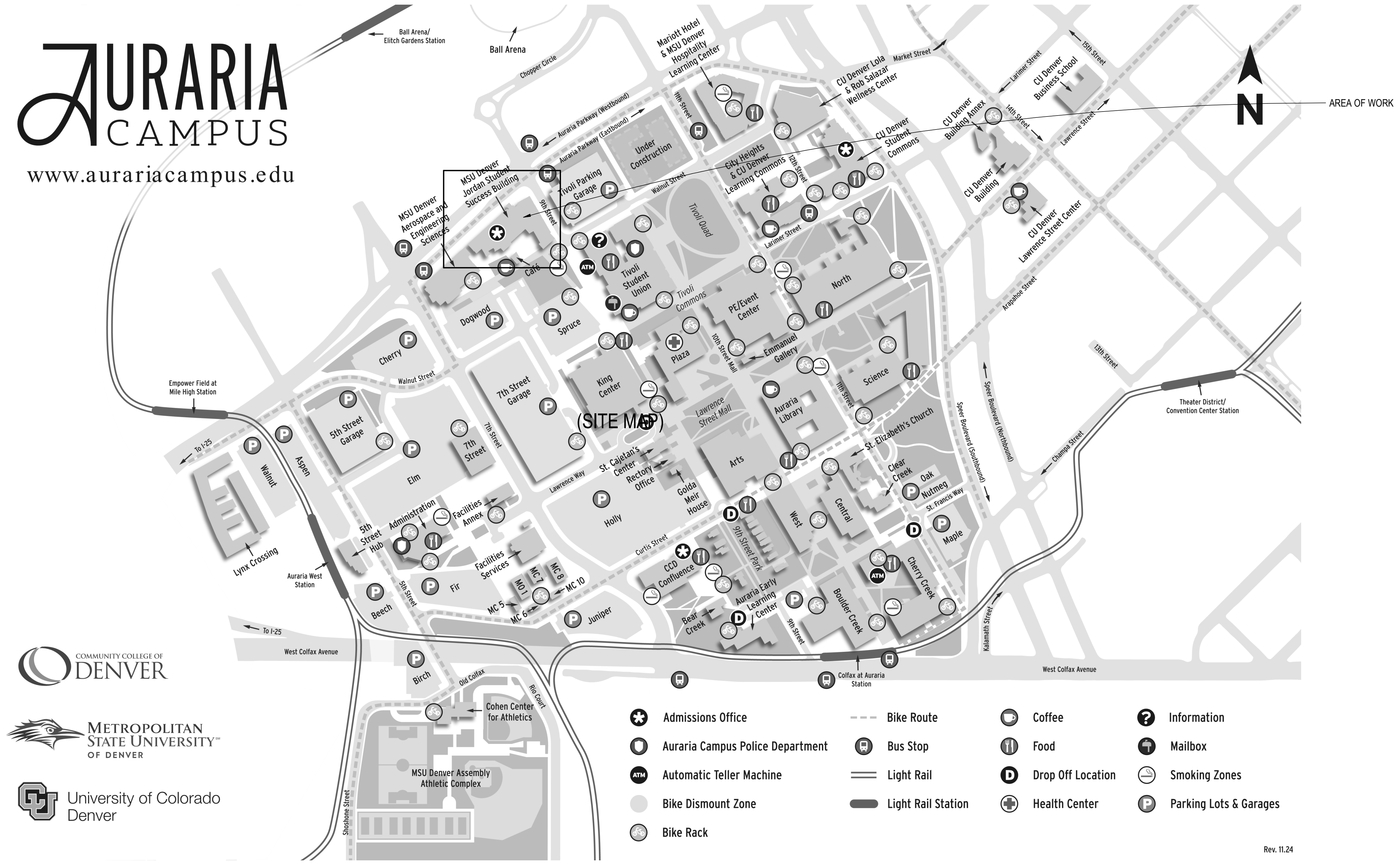
METRO STATE UNIVERSITY

CAVEA LIGHTING UPGRADE

JULY 17, 2026

100% CONSTRUCTION DOCUMENTS

PROJECT LOCATION: 890 AURARIA PKWY, DENVER, CO 80204



	GENERAL
G-000	PROJECT COVER SHEET
	ELECTRICAL
E-001	ELECTRICAL LEGEND
E-002	ELECTRICAL ONE-LINE DIAGRAM
E-114	PARTIAL LIGHTING PLAN LEVEL 4
E-601	ELECTRICAL SCHEDULES
E-602	LIGHTING COMCHECK

RMH GROUP
making a difference through engineering SM

METROPOLITAN STATE UNIVERSITY OF DENVER

COLORADO LICENSED PROFESSIONAL ENGINEER
SHARA RENEE JOHNSON
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ELECTRICAL LEGEND

(NOTE: NOT ALL SYMBOLS SHOWN ARE USED ON THESE DRAWINGS)

- ONE LINE SYMBOLS -

SYMBOL	DESCRIPTION
⌒	CIRCUIT BREAKER
≪⌒≫	DRAW-OUT CIRCUIT BREAKER (MOLDED INSULATED CASE)
≪52≫	DRAW-OUT POWER CIRCUIT BREAKER
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	FUSE WITH SWITCH
	SWITCH
PANEL XXX	PANELBOARD
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▭	GROUND BUS
	WEATHERHEAD
Ⓧ	MOTOR
△	DELTA CONNECTION
Y	WYE CONNECTION
	GROUNDED WYE CONNECTION
	GROUNDED WYE CONNECTION WITH RESISTOR GROUND
	GROUNDED WYE CONNECTION WITH REACTOR GROUND
Ⓜ	METERING DEVICE
	CURRENT TRANSFORMER
	POTENTIAL TRANSFORMER
	LOAD-BREAK CONNECTOR
##	PROTECTIVE RELAY DEVICE
Ⓚ	KEY INTERLOCK
2400V, 100A 10SEC. RESISTOR	RESISTOR
	CONTACT NORMALLY OPEN
	CONTACT NORMALLY CLOSED
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	SINGLE BATTERY
	MULTIPLE BATTERIES
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	THERMAL ELEMENT, OVERLOAD RELAY

- GENERAL -

SYMBOL	DESCRIPTION
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	CONDUIT BREAK SYMBOL
	CONDUIT CAP
	CONDUIT CHANGE IN ELEVATION
—●	CONDUIT STUB DOWN (OUT OF DRAWING LIMITS)
—○	CONDUIT STUB UP (OUT OF DRAWING LIMITS)
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Ⓙ	WALL MOUNTED JUNCTION BOX
Ⓙ	FLOOR MOUNTED JUNCTION BOX
	PUSH BUTTON A = ABORT DA = DURESS ALARM EPO = EMERGENCY POWER OFF IC = INTERCOM ST = SHUNT TRIP
$S^{X}_{\#}$	SWITCH SYMBOL (#) SINGLE POLE (IF BLANK) 2 = DOUBLE POLE 3 = THREE-WAY 4 = FOUR-WAY AS = ADJUSTABLE SPEED D = DIMMER K = KEY OPERATED LV = LOW VOLTAGE M = MANUAL MOTOR SWITCH OS = OCCUPANCY SENSOR P = WITH PILOT LIGHT T = TIMER TO = THERMAL OVERLOAD VS = VACANCY SENSOR WP = WEATHERPROOF x = SMALL LETTER - LUMINAIRES CONTROLLED XP = EXPLOSION PROOF
S S	DUAL SWITCH
$	INTERCOM SWITCH

- FIRE ALARM DEVICES -

SYMBOL	DESCRIPTION
B	BELL
D D	DUCT SMOKE DETECTOR
	FIRE FIGHTER'S TELEPHONE JACK
H	COMBINATION FIRE HORN/STROBE LIGHT
S	COMBINATION FIRE SPEAKER/STROBE LIGHT
	FIRE ALARM STROBE LIGHT
	FIRE ALARM STROBE, CEILING MOUNT
Ⓢ	FIRE ALARM STROBE/SPEAKER, CEILING MOUNT
H	FIRE HORN
H	DUAL PROJECTION FIRE HORN
F	MANUAL PULL STATION
■	MAGNETIC DOOR HOLD OPEN
Ⓓ F ⬡ X	DETECTOR / DETECTOR UNDER FLOOR F = FLAME I = IONIZATION TYPE P = PHOTOELECTRIC TYPE T = THERMAL TYPE
Ⓑ	BEAM DETECTOR
	PULL STATION/TELEPHONE JACK
Ⓢ	DELUGE VALVE

- SPECIAL SYSTEMS DEVICES -

SYMBOL	DESCRIPTION
▽	DATA OUTLET
▽	COMBINATION TELEPHONE/DATA OUTLET
▼	TELEPHONE OUTLET
	TELEVISION JACK
	CEILING MOUNTED DATA OUTLET
	CEILING MOUNTED TELEPHONE/DATA OUTLET
	CEILING MOUNTED TELEPHONE OUTLET
	FLOOR MOUNTED DATA OUTLET
	FLOOR MOUNTED TELEPHONE/DATA OUTLET
	FLOOR MOUNTED TELEPHONE OUTLET
TTB	TELEPHONE TERMINAL BOARD
DTB	DATA TERMINAL BOARD
M	MICROPHONE OUTLET
Ⓢ X	SPEAKER V = WITH INTEGRAL VOLUME CONTROL
S	WALL MOUNTED SPEAKER
V	WALL MOUNTED VOLUME CONTROL
Ⓒ	CLOCK RECEPTACLE OUTLET
	SECURITY CAMERA
Ⓣ	THERMOSTAT
XX	SPECIAL SYSTEMS SYMBOLS: AB = ABORT BUTTON AMP = AMPLIFIER ANN = ANNUNCIATOR ASD = ADJUSTABLE SPEED DRIVE ATS = AUTOMATIC TRANSFER SWITCH B = BUZZER BMS = BALANCED MAGNETIC STRIP CAM = CONTROL ADDRESSABLE MODULE CH = CHIME CK = CARD READER/KEYPAD CR = CARD READER DA = DURESS ALARM PUSH-BUTTON DC = DOOR CONTACT DE = DELAYED EGRESS DMA = DOOR MANAGEMENT ALARM DO = DOOR OPERATOR DS = DOOR STRIKE DTC = DATA TERMINAL CABINET DVR = DIGITAL VIDEO RECORDER EC = ELECTRICAL CONTACTOR EL = ELECTRIC LOCK ES = ELECTRIC STRIKE F = MANUAL PULL STATION FAA = FIRE ALARM ANNUNCIATOR FACP = FIRE ALARM CONTROL PANEL FS = FLOW SWITCH FSD = FIRE SMOKE DAMPER GB = GLASSBREAK DETECTOR HC = HANDICAP PUSH-BUTTON IC = INTERCOM CALL STATION IP = INDICATOR PANEL KP = KEYPAD MAM = MONITOR ADDRESSABLE MODULE MH = MANHOLE MD = MOTORIZED DAMPER MS = MOTION SENSOR ML = MAGNETIC LOCK NCMS = NURSE CALL MASTER STATION CM-CENTRALIZED PHYSIOLOGICAL MONITORING STATION NCT = NURSE CALL TERMINAL CABINET OS = OCCUPANCY SENSOR PA = PUBLIC ADDRESS PB = PULLBOX PIR = PASSIVE INFRARED DETECTOR PM = POWER QUALITY METER PS = PRESSURE SWITCH R = REMOTE INDICATING LIGHT RT = REMOTE TEST STATION RX = REQUEST TO EXIT PUSH-BUTTON S = WALL MOUNTED SPEAKER SAP = SECURITY ALARM PANEL SPD = SURGE PROTECTIVE DEVICE TC = TIME CLOCK TS = TAMPER SWITCH VFD = VARIABLE FREQUENCY DRIVE VM = VIDEO MONITOR VS = VACANCY SENSOR

- POWER -

SYMBOL	DESCRIPTION
	DUPLEX RECEPTACLE D = DEDICATED CIRCUIT IG = ISOLATED GROUND DEVICE ⏻ = AUTOMATICALLY CONTROLLED RECEPTACLE
GFI	GFI = GROUND FAULT CIRCUIT INTERRUPTER
	FLOOR MOUNTED DUPLEX RECEPTACLE
	FLOOR MOUNTED FOURPLEX RECEPTACLE
	FLOOR MOUNTED SPECIAL PURPOSE RECEPTACLE
	CEILING MOUNTED DUPLEX RECEPTACLE
	CEILING MOUNTED FOURPLEX RECEPTACLE
	CEILING MOUNTED SPECIAL PURPOSE RECEPTACLE
	FOURPLEX RECEPTACLE
	RANGE RECEPTACLE
	SINGLE RECEPTACLE
	SPECIAL PURPOSE RECEPTACLE
	SWITCHED RECEPTACLE
	DUPLEX EMERGENCY/CRITICAL
	FOURPLEX EMERGENCY/CRITICAL
	DUPLEX-2PORT USB
	4PORT USB
	DISCONNECT SWITCH
	FUSED DISCONNECT SWITCH
	ENCLOSED CIRCUIT BREAKER
⊠	MOTOR STARTER
⊠	COMBINATION MOTOR STARTER
S	COMBINATION SWITCH AND RECEPTACLE
⋈	CONDUIT SEAL OFF
⊙	FIRE RATED POKE-THROUGH
	PARTITION CIRCUIT SPLIT
	POWER POLE
W	SURFACE RACEWAY
M	METER

- NURSE CALL -

SYMBOL	DESCRIPTION
CB	CODE BLUE STATION
CP	CODE PINK STATION
EP	EMERGENCY CALL PULL STATION
SP	SINGLE PATIENT STATION
SA	STAFF ASSIST STATION
AI	AUXILIARY INPUT STATION
DS	DUTY STATION
SS	STAFF STATION
BI	BED INTERFACE STATION
2B	TWO BUTTON STATION
BM	BED MANAGEMENT STATION
D	DOME LIGHT
Z	ZONE LIGHT INTERSECTION
RL	REMOTE LOCATOR
PM	PATIENT MONITOR

- LIGHTING - (REFER TO LUMINAIRE SCHEDULE)

SYMBOL	DESCRIPTION
X* #	LUMINAIRES X = FIXTURE DESIGNATION # = BRANCH CIRCUIT NUMBER * = SWITCH LEG IDENTIFIER
	SHADING INDICATES LUMINAIRE ON LIFE SAFETY
	SHADING INDICATES PORTION OF LUMINAIRE ON LIFE SAFETY
	WALL MOUNTED LUMINAIRE
	STRIP LIGHT
	STRIP LIGHT WITH LIFE SAFETY
	POLE MOUNTED LUMINAIRE (QUANTITY OF LUMINAIRES PER POLE AS INDICATED ON PLANS)
○ □	DOWNLIGHT LUMINAIRE
◑	WALL WASHER LUMINAIRE
	ADJUSTABLE LUMINAIRE
	PENDANT LUMINAIRE
	TRACK LIGHTING
Ⓟ	PHOTOCELL
	EXIT LIGHT (WITH FACES AND DIRECTION ARROWS INDICATED)
	WALL MOUNTED EXIT LIGHT (WITH FACES AND DIRECTION ARROWS INDICATED)
	WALL MOUNTED BATTERY PACK EMERGENCY LIGHT

- GROUNDING SYMBOLS -

SYMBOL	DESCRIPTION
⊗	GROUND TEST WELL
—	GROUNDING CONDUCTOR
⦿	LIGHTNING PROTECTION AIR TERMINAL
—●	BONDING POINT
G	GROUND BAR
⏚	ELECTRICAL GROUND
	GROUND ROD
	GROUND ROD WITH INSPECTION TEST WELL
	PIGTAIL

- EQUIPMENT -

SYMBOL	DESCRIPTION
	DISTRIBUTION PANEL
	EXISTING DISTRIBUTION PANEL
	NEW PANEL, FLUSH MOUNTED
	EXISTING PANEL, FLUSH MOUNTED
	NEW PANEL, SURFACE MOUNTED
	EXISTING PANEL, SURFACE
T	TRANSFORMER

APPLICABLE CODES

AHJ: STATE OF COLORADO

FIRE AUTHORITY: DENVER FIRE

REMODEL ☒ NEW ☐

YEAR	CODE
2024	INTERNATIONAL BUILDING CODE
2024	INTERNATIONAL MECHANICAL CODE
2024	INTERNATIONAL PLUMBING CODE
2021	INTERNATIONAL ENERGY CONSERVATION CODE
2024	INTERNATIONAL FIRE CODE
2023	NATIONAL ELECTRICAL CODE
2025	DENVER BUILDING AND FIRE CODES

IS THE BUILDING FULLY SPRINKLERED? ☐ Yes ☒ No

- GRAPHIC SYMBOLS -

SYMBOL	DESCRIPTION
1	KEY NOTE
#	REVISION NUMBER
X	DETAIL NOTE
X	X = DENOTES ALL LUMINAIRES IN THE RESPECTIVE AREA ARE THE TYPE INDICATED, REFER TO LUMINAIRE SCHEDULE
TAG	PANEL FLAG
TAG	OWNER EQUIPMENT TAG
S# X	LIGHTING CONTROL SEQUENCE INDICATION, SEE LIGHTING CONTROL SEQUENCE OF OPERATION SCHEDULE FOR INFORMATION X = DENOTES ILLUMINATION SET POINT
TAG	MECHANICAL EQUIPMENT TAG
▭	SHADING INDICATES EQUIPMENT
▨	HATCHING INDICATES ITEM(S) TO BE REMOVED
A100	ROOM NUMBER
N	NORTH ARROW
1 E2.1	DETAIL BUBBLE DETAIL NUMBER SHEET NUMBER - WHERE DETAIL IS SHOWN
2 E2.2	SECTION CUT SECTION NUMBER/LETTER SHEET NUMBER - WHERE SECTION IS SHOWN

- ABBREVIATIONS -

(E)	EXISTING	IDF	INTERMEDIATE DISTRIBUTION FRAME
(N)	NEW	IG	ISOLATED GROUND
(PART)	PARTIAL CIRCUIT	JB	JUNCTION BOX
A	AMP	LED	LIGHT EMITTING DIODE
AC	ABOVE COUNTER TOP	LTG	LIGHTING
AF	AMP FRAME, AMP FUSE	LV	LOW VOLTAGE
AFF	ABOVE FINISHED FLOOR	MATV	MASTER ANTENNA T.V.
AFG	ABOVE FINISHED GRADE	MCB	MAIN CIRCUIT BREAKER
AHJ	AUTHORITY HAVING JURISDICTION	MCC	MOTOR CONTROL CENTER
AIC	AMPS INTERRUPTING CAPACITY	MDF	MAIN DISTRIBUTION FRAME
AL	ALUMINUM	MH	MANHOLE
APL	APPLIANCE	MLO	MAIN LUGS ONLY
AS	AMP SWITCH	MOP	METHOD OF PROCEDURE
AT	AMP TRIP	MTD	MOUNTED
ATS	AUTOMATIC TRANSFER SWITCH	MTG	MOUNTING
AV	AUDIOVISUAL	MTR	MOTOR
BJ	BONDING JUMPER	MV	MEDIUM VOLTAGE
C	CONDUIT	N	NEUTRAL
CB	CIRCUIT BREAKER	NC	NORMALLY CLOSED
CCT	CIRCUIT	NEC	NATIONAL ELECTRICAL CODE
CCTV	CLOSED CIRCUIT TV	NF	NON-FUSED
CLG	CEILING	NIC	NOT IN CONTRACT
CU	COPPER	NL	NIGHT LIGHT
DISC	DISCONNECT	NO	NORMALLY OPEN
DIST	DISTRIBUTION	OHE	OVERHEAD ELECTRIC
E/G	ENGINE/GENERATOR	P	POLE
ELEC	ELECTRIC, ELECTRICAL	PA	PUBLIC ADDRESS
ELEV	ELEVATOR ELEVATION	PB	PULL BOX
EM	EMERGENCY	PH	PHASE
EQPT	EQUIPMENT	PNL	PANEL
F	FUSE	PWR	POWER
FA	FIRE ALARM	RCPT	RECEPTACLE
FAA	FIRE ALARM ANNUNCIATOR	SBC	STRANDED BARE COPPER
FACP	FIRE ALARM CONTROL PANEL	SPD	SURGE PROTECTIVE DEVICE
FIXT	FIXTURE	SW	SWITCH
FO	FIBER OPTIC	TEL	TELEPHONE
FLR	FLOOR	TP	TAMPERPROOF
FLUOR	FLUORESCENT	TV	TELEVISION
G	GROUND (EQUIPMENT)	UF	UNDERFLOOR
GEN	GENERATOR	UNO	UNLESS NOTED OTHERWISE
GFI	GROUND FAULT CIRCUIT INTERRUPTER	UPS	UNINTERRUPTIBLE POWER SUPPLY
GF	GROUND FAULT PROTECTION	V	VOLTAGE
GND	GROUND	VFD	VARIABLE FREQUENCY DRIVE
HC	HANDICAPPED	VP	VAPOR PROOF
HG	HOSPITAL GRADE	W	WIRE
HP	HORSEPOWER	WAP	WIRELESS ACCESS POINT
HV	HIGH VOLTAGE	WG	WIRE GUARD
INC	INCANDESCENT	WP	WEATHERPROOF
		XFMR	TRANSFORMER
		XP	EXPLOSION PROOF

REV: 10/21/2024

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GENERAL PROJECT NOTES

1. LIGHT LINE WEIGHT INDICATES EXISTING. HEAVY LINE WEIGHT INDICATES NEW CONSTRUCTION.
2. SPECIFICATIONS ARE A PART OF THE CONSTRUCTION DOCUMENTS. SHOULD ANY CONFLICT ARISE BETWEEN THE DRAWINGS AND SPECIFICATIONS, BRING SUCH CONFLICT TO THE ATTENTION OF THE ENGINEER FOR RESOLUTION. UNLESS OTHERWISE DIRECTED BY ENGINEER, THE MOST STRINGENT REQUIREMENT WILL PREVAIL.
3. DATA ON THE DRAWINGS IS AS EXACT AS COULD BE REASONABLY SECURED. ABSOLUTE ACCURACY IS NOT GUARANTEED. VERIFY EXACT LOCATIONS, MEASUREMENTS, LEVELS, SPACE REQUIREMENTS, POTENTIAL CONFLICTS WITH OTHER TRADES. ADAPT WORK TO ACTUAL CONDITIONS AT THE SITE. VISIT THE SITE TO BECOME THOROUGHLY FAMILIAR WITH THE ACTUAL CONDITIONS OF THIS PROJECT BEFORE SUBMITTING COSTS. THESE DRAWINGS ARE DIAGRAMMATIC IN NATURE; DO NOT SCALE. THESE DRAWINGS DO NOT SHOW MATERIALS FOR A COMPLETE INSTALLATION; PLAN THE INSTALLATION AND LAYOUT OF THE WORK AS DIAGRAMMED IN THESE DOCUMENTS. REFER TO FLOOR PLANS, SCHEMATICS AND DIAGRAMS OF OTHER TRADES FOR ELECTRICAL REQUIREMENTS, BRANCH CIRCUITS AND OTHER ELECTRICAL CONNECTIONS NOT INDICATED ON THESE DOCUMENTS.
4. FIRE-SEAL ALL PENETRATIONS THROUGH RATED ASSEMBLIES WITH MATERIALS CAPABLE OF PREVENTING THE PASSAGE OF FLAMES AND HOT GASES WHEN SUBJECTED TO THE REQUIREMENTS OF THE TEST STANDARD SPECIFIC FOR FIRE STOPS ASTM E814.
5. JUNCTION BOXES FOR LUMINAIRES AND OUTLETS ARE NOT INDICATED. PROVIDE THE PROPER QUANTITY OF JUNCTION BOXES TO MEET LOCAL CODE AND NATIONAL ELECTRICAL CODE.
6. EXECUTE THE WORK IN ACCORDANCE WITH SUPPORTING OBJECTS FOR SEISMIC ZONE REQUIRED BY STATE AND LOCAL CODES FOR ALL CEILING ATTACHED OBJECTS AND FLOOR ATTACHED EQUIPMENT INCLUDING, BUT NOT LIMITED TO: PENDANT LIGHTING FIXTURES, GENERAL LIGHTING, MULTIPLE RACEWAYS, GENERATOR, TRANSFORMER, ELECTRICAL SWITCHGEAR, SWITCHBOARDS AND OTHER ELECTRICAL EQUIPMENT.
7. COORDINATE AND SCHEDULE ALL ELECTRICAL OUTAGES AND OUTAGE TIME FRAMES WITH OWNER'S REPRESENTATIVE AT LEAST TWO WEEKS PRIOR TO OUTAGE. ACCOMPLISH ALL PREP WORK PRIOR TO OUTAGE TO MINIMIZE THE OUTAGE TIME FRAME. SUBMIT A DETAILED METHOD OF PROCEDURE (MOP) FOR ALL OUTAGES FOUR WEEKS PRIOR TO OUTAGE. PERFORM OUTAGES DURING OFF HOURS AT A TIME ACCEPTABLE BY THE OWNER.
8. UPDATE PANEL SCHEDULES TO REFLECT INSTALLED CONDITIONS AFTER CONSTRUCTION IS COMPLETE.
9. UNLESS OTHERWISE INDICATED, ALL CONDUCTORS FOR BRANCH CIRCUITS SHALL BE #12 AWG PROTECTED BY 20-AMPERE CIRCUIT BREAKERS. INCREASE CONDUCTOR SIZE TO ACCOUNT FOR VOLTAGE DROP FOR ALL 120-VOLT CIRCUITS OVER 75 FEET, AND ALL 277-VOLT CIRCUITS OVER 150 FEET TO THE FIRST OUTLET. CONDUCTOR SIZE SHALL BE UNIFORM FOR THE ENTIRE LENGTH OF THE CIRCUIT UNLESS NOTED OTHERWISE. HOMERUNS WHICH INDICATE UPGRADING CIRCUIT CONDUCTORS FOR VOLTAGE DROP, E.G., #10AWG WIRE ON 20-AMPERE CIRCUIT, SHALL HAVE THE CONDUCTOR SIZE INDICATED CARRIED THROUGHOUT THE CIRCUIT TO ALL JUNCTION BOXES UP TO AND INCLUDING THE J-BOX NEAREST THE LAST DEVICE OR LUMINAIRE.
10. THESE DRAWINGS HAVE BEEN PREPARED BASED ON INFORMATION PROVIDED BY OTHERS. DATA PRESENTED ON THIS DRAWING IS AS ACCURATE AS CAN BE DETERMINED, BUT ACCURACY IS NOT GUARANTEED. WHILE THIS INFORMATION IS BELIEVED TO BE RELIABLE, THE ENGINEER IS NEITHER RESPONSIBLE FOR ITS ACCURACY, NOR ERRORS NOR OMISSIONS WHICH MAY HAVE BEEN INCORPORATED INTO THESE DOCUMENTS. FIELD VERIFICATION OF ALL AFFECTED COMPONENTS IS REQUIRED.
11. PATCH AND PAINT ANY DAMAGED SURFACES DUE TO DEMOLITION AND CONSTRUCTION TO MATCH EXISTING CONDITIONS. SEAL ALL PENETRATIONS THROUGH RATED CEILINGS AND WALLS WITH UL LISTED SEALANTS AND FIRESTOPPING MATERIALS.
12. UP-TO-DATE ELECTRICAL RECORD DRAWINGS ARE NOT AVAILABLE FOR THIS PROJECT. INFORMATION FOR EXISTING CIRCUITRY IS BASED ON EXISTING PANEL DIRECTORIES, AVAILABLE DRAWINGS, AND ASSUMPTIONS. LOCATIONS AND INFORMATION FOR EXISTING ELECTRICAL DEVICES AND EQUIPMENT SHOWN ON THESE DOCUMENTS ARE APPROXIMATE AND WERE DERIVED FROM FIELD OBSERVATION AND AVAILABLE RECORD DRAWINGS. VERIFY ACTUAL FIELD CONDITIONS PRIOR TO STARTING WORK.

DEMOLITION PROJECT NOTES

1. WORK SHOWN HATCHED IS EXISTING TO BE REMOVED. REMOVE ALL DEVICES ON WALLS WHICH ARE REMOVED. WHERE DEVICES ARE REMOVED, REMOVE ASSOCIATED CIRCUIT CONDUCTORS AND CONDUIT BACK TO THE SERVING PANELBOARD OR NEXT DEVICE ON THE CIRCUIT WHICH REMAINS (MAINTAIN CIRCUIT CONTINUITY).
2. UNLESS OTHERWISE INDICATED, DEMOLITION WASTE BECOMES PROPERTY OF CONTRACTOR. REMOVE DEMOLITION WASTE MATERIALS FROM PROJECT SITE AND DISPOSE OF THEM IN AN APPROVED CONSTRUCTION AND DEMOLITION WASTE LANDFILL ACCEPTABLE TO AUTHORITIES HAVING JURISDICTION. COORDINATE DISPOSAL LOCATION WITH OWNERS REPRESENTATIVE PRIOR TO DISPOSING OF ANY REMOVED ELECTRICAL EQUIPMENT.
3. EXISTING CIRCUIT INFORMATION SHOWN ON THE PLANS IS FOR REFERENCE ONLY. TRACE ALL EXISTING CIRCUITS AFFECTED BY THIS WORK TO CONFIRM SOURCE PANEL AND CIRCUIT NUMBERS.
4. INFORMATION ON THE DRAWINGS HAS BEEN TAKEN FROM RECORD DRAWINGS. DEMOLITION DRAWINGS ARE INTENDED TO GIVE A COMMON BASIS FOR WORK. VISIT THE SITE, PRIOR TO FINAL PRICING AND BECOME FAMILIAR WITH THE EXTENT OF WORK REQUIRED. NO EXTRAS WILL BE ALLOWED FOR ALTERATIONS OF A FORESEEABLE NATURE REQUIRED TO ACHIEVE THE END RESULT AS INDICATED BY THE CONTRACT DOCUMENTS.
5. MAINTAIN FIRE ALARM SYSTEM OPERATION IN ALL AREAS OUTSIDE OF THE AREA OF WORK DURING CONSTRUCTION.

LIGHTING PROJECT NOTES

1. EXISTING CIRCUIT INFORMATION SHOWN ON THE PLANS IS FOR REFERENCE ONLY. TRACE ALL EXISTING CIRCUITS AFFECTED BY THIS WORK TO CONFIRM SOURCE PANEL AND CIRCUIT NUMBERS.
2. UPPERCASE LETTER DENOTES LUMINAIRE TYPE. LOWERCASE LETTER INDICATES CONTROL ZONE.
3. EMERGENCY LUMINAIRES SHALL BE SWITCHED WITH THE REST OF THE LUMINAIRES IN THE ROOM OR AS INDICATED ON THE DRAWINGS BY LIGHTING CONTROL ZONE. PROVIDE ONE EMERGENCY TRANSFER RELAY PER CONTROL ZONE. RELAY SHALL BE INSTALLED IN AN ACCESSIBLE LOCATION.
4. TO ACHIEVE THE INDICATED SEQUENCE OF OPERATION, PROVIDE ALL COMPONENTS AND DIMMING CABLING REQUIRED FOR THE LIGHTING CONTROL SYSTEM, WHETHER INDICATED OR NOT. REFER TO LIGHTING CONTROL SEQUENCE OF OPERATION SCHEDULE FOR ADDITIONAL INFORMATION.
5. PROVIDE DIMMERS COMPATIBLE WITH LED DRIVERS AND POWER SUPPLIES PER MANUFACTURER'S RECOMMENDATIONS. PROVIDE COMPONENTS AND ACCESSORIES TO ACCOMPLISH COMPLETE DIMMING FUNCTIONALITY AS SHOWN ON PLANS AND NOTED IN THE LUMINAIRE SCHEDULE.
6. FOR LIGHT DIMMING BRANCH CIRCUITS, PROVIDE A DEDICATED NEUTRAL CONDUCTOR FOR EACH BRANCH CIRCUIT CURRENT CARRYING CONDUCTOR. CIRCUITS MAY BE COMBINED INTO HOMERUNS OF UP TO SIX CURRENT CARRYING CONDUCTORS WITH PROPER DE-RATING AS REQUIRED BY NEC. PROVIDE FULL LENGTH STRIPING FOR DEDICATED NEUTRALS TO MATCH ASSOCIATED CURRENT CARRYING CONDUCTOR COLOR.
7. LIGHTING CONTROLS COMMISSIONING SHALL COMPLY WITH IECC 2024, INCLUDING BUT NOT LIMITED TO COMMISSIONING DOCUMENTS, SYSTEM FUNCTIONAL TESTING, AND RECORD DOCUMENTS. REFER TO SPECIFICATIONS FOR ADDITIONAL INFORMATION.

DESCRIPTION	REV	DATE

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07.17.2026

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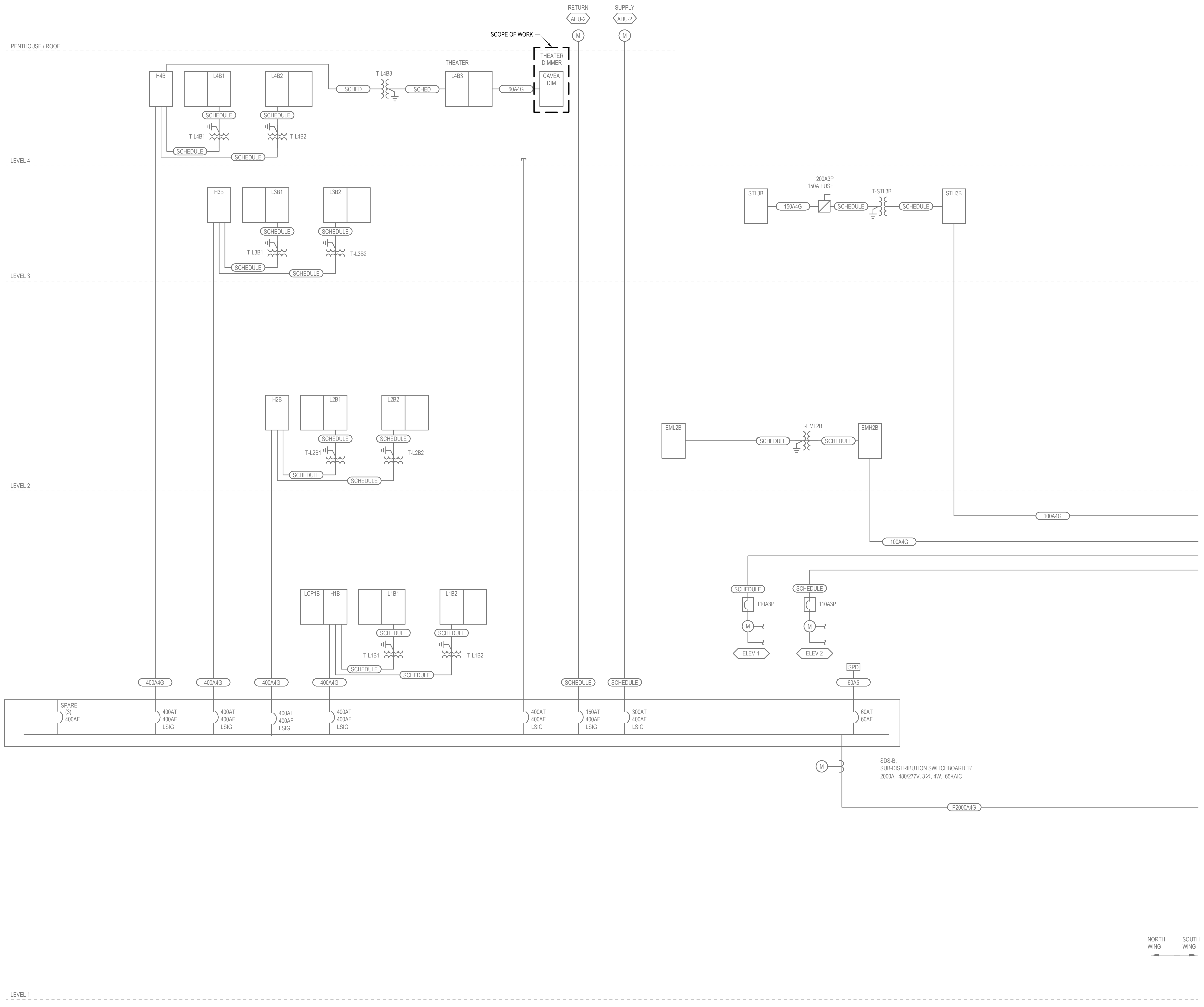
METRO STATE
UNIVERSITY

CAVEA LIGHTING UPGRADE
ELECTRICAL LEGEND

Date: JULY 17, 2026
Scale: None
Designed By: SRJ
Drawn By: MLB
Approved By: SKB
Project No: 21607
Project Status: 100% CD
RMH-21607

SHT NO: E-001
REVISION



1
E-002
PARTIAL ONE-LINE DIAGRAM - FOR REFERENCE ONLY
SCALE: NONE

SHEET NOTES

1. COORDINATE AND SCHEDULE ALL ELECTRICAL OUTAGES AND OUTAGE TIME FRAMES WITH OWNER'S REPRESENTATIVE AT LEAST TWO WEEKS PRIOR TO OUTAGE. ACCOMPLISH ALL PREP WORK PRIOR TO OUTAGE TO MINIMIZE THE OUTAGE TIME FRAME. SUBMIT A DETAILED METHOD OF PROCEDURE (MOP) FOR ALL OUTAGES FOUR WEEKS PRIOR TO OUTAGE. PERFORM OUTAGES DURING OFF HOURS AT A TIME ACCEPTABLE BY THE OWNER.

2. PROVIDE TEMPORARY POWER FOR SMALL TOOLS AND TASK LIGHTING.

3. COMPLETELY REMOVE ALL UNUSED CONDUCTORS AND RACEWAYS BACK TO SOURCE. MARK UNUSED CIRCUIT BREAKERS AND FUSED SWITCHES THAT REMAIN AS "SPARE". MAINTAIN CONTINUITY OF CIRCUITS FOR ANY PORTIONS OF CIRCUITS TO REMAIN.

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UNIVERSITY

CAVEA LIGHTING UPGRADE
ELECTRICAL ONE-LINE
DIAGRAM

Date: JULY 17, 2026
Scale: NONE
Designed By: SRJ
Drawn By: MLB
Approved By: SKB
Project No: 21607
Project Status: 100% CD
RMH-21607

SHT NO:
E-002
REVISION

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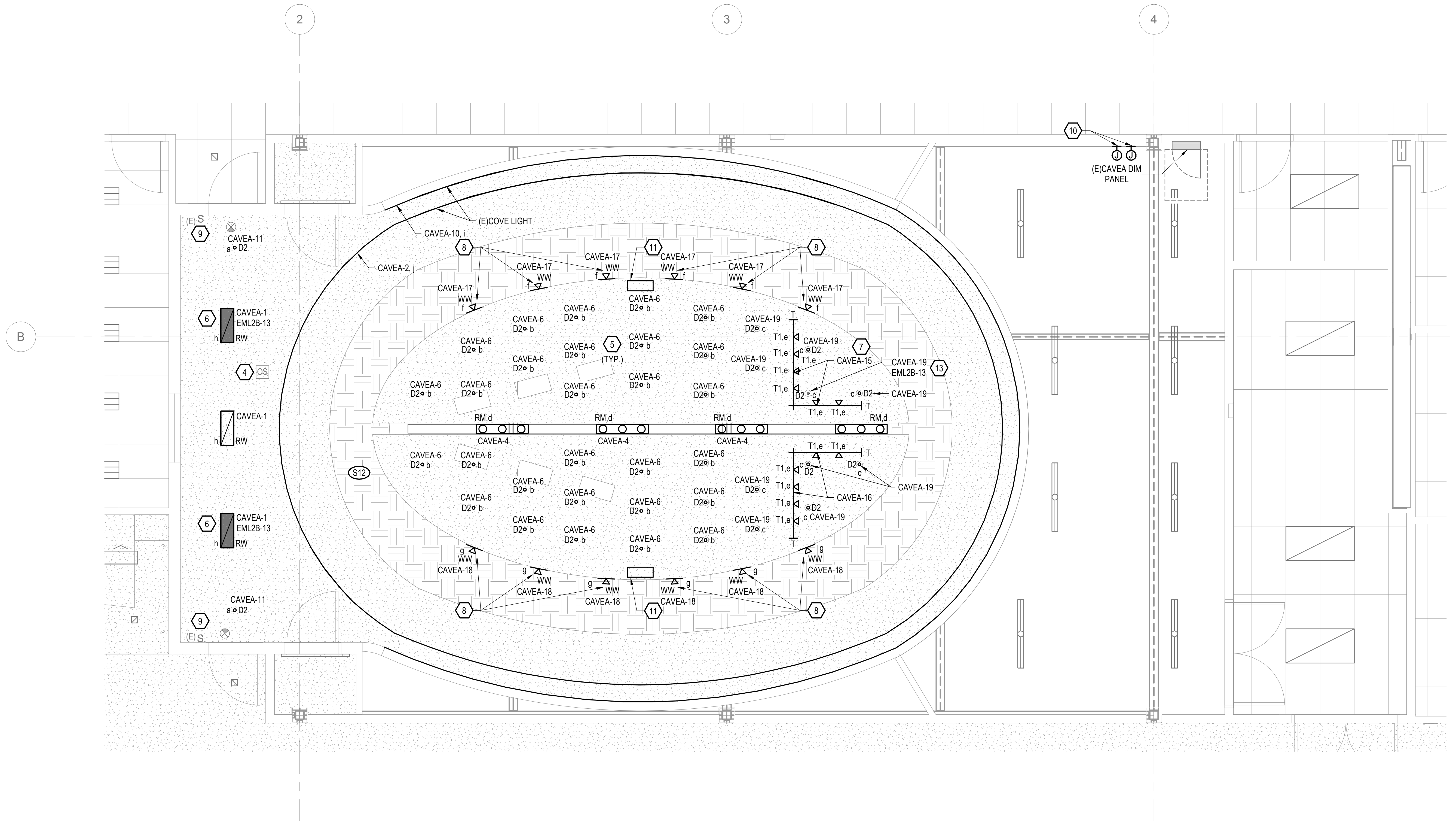
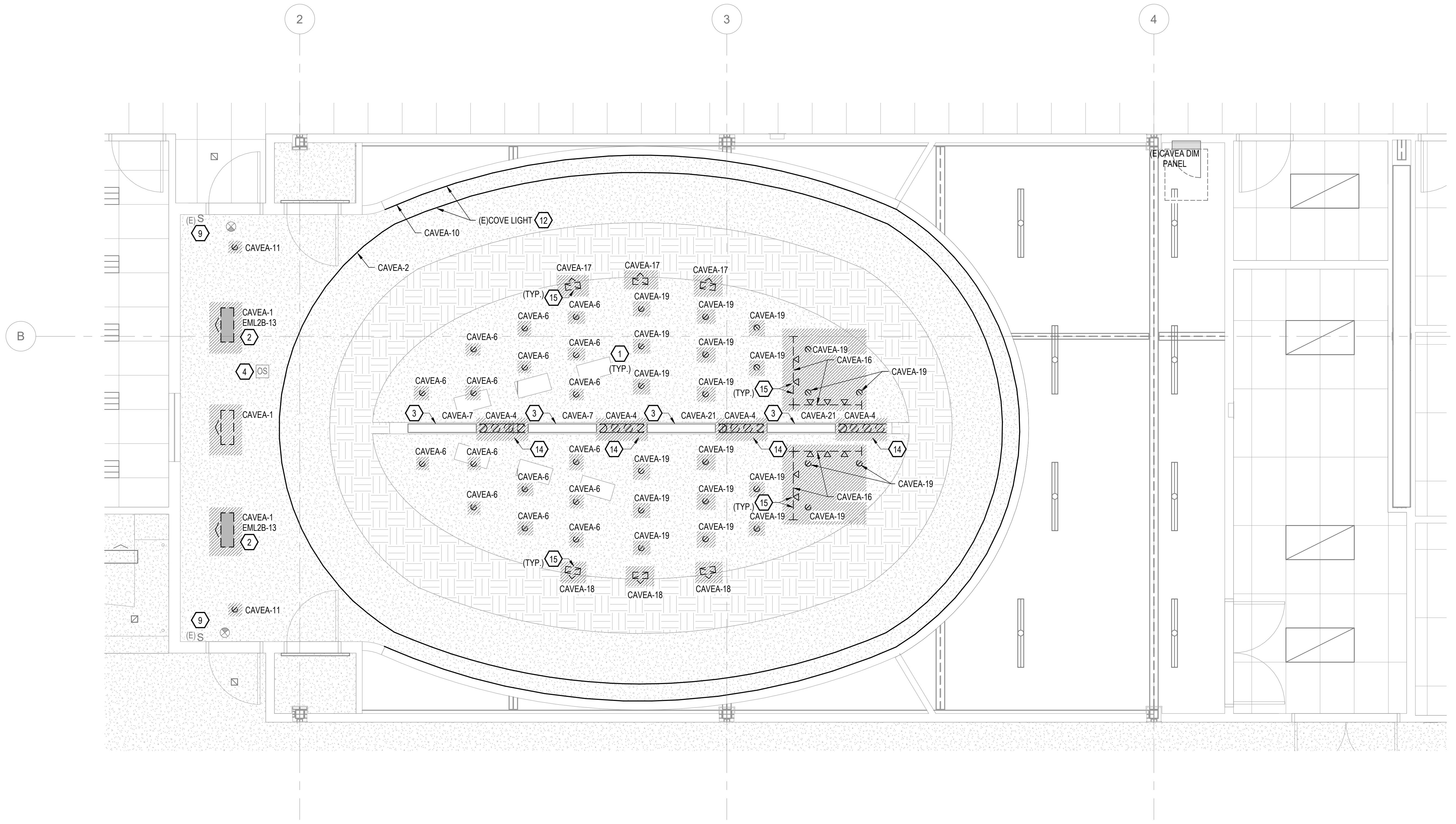


1 PARTIAL LIGHTING INSTALLATION PLAN LEVEL 04

SCALE: 1/4" = 1'-0"



2 PARTIAL LIGHTING DEMOLITION PLAN LEVEL 04

E-114

SCALE: NONE

KEY NOTES

1. DISCONNECT AND REMOVE EXISTING LUMINAIRES WITHIN AREA OF WORK. PRESERVE AND PROTECT EXISTING BRANCH CIRCUITS FOR CONNECTION TO NEW LUMINAIRES.
2. DISCONNECT AND REMOVE EXISTING LUMINAIRE. PRESERVE AND PROTECT EXISTING BRANCH CIRCUIT AND TRANSFER RELAY FOR CONNECTION TO NEW LUMINAIRE.
3. REMOVE EXISTING FLUORESCENT LAMP AND DISCONNECT EXISTING BRANCH CIRCUIT FROM LUMINAIRE. LUMINAIRE SHALL BE ABANDONED IN PLACE.
4. PRESERVE AND PROTECT EXISTING OCCUPANCY SENSOR. REPROGRAM EXISTING SENSOR AS DESCRIBED IN THE LIGHTING CONTROL SEQUENCE OF OPERATION SCHEDULE. IF EXISTING SENSOR IS UNABLE TO BE REPROGRAMMED, REPLACE EXISTING SENSOR.
5. CONNECT NEW LUMINAIRES WITHIN AREA OF WORK TO EXISTING BRANCH CIRCUITS SALVAGED DURING DEMOLITION. PROVIDE ADDITIONAL CONDUIT AND WIRING TO EXTEND BRANCH CIRCUITS WHERE REQUIRED.
6. CONNECT NEW LUMINAIRE TO EXISTING BRANCH CIRCUITS AND TRANSFER RELAY SALVAGED DURING DEMOLITION. RELAY SWITCHES LUMINAIRE TO EMERGENCY CIRCUIT UPON LOSS OF NORMAL POWER.
7. PROVIDE NEW BRANCH CIRCUIT TO NEW TRACK LIGHTING SYSTEM.
8. SURFACE MOUNT WALL WASH LUMINAIRE ON TOP OF LOWER OVAL CEILING AND SPACE EQUALLY BETWEEN LUMINAIRES, MAXIMIZE SPACING SO THAT LUMINAIRE MAINTAINS ADJUSTABILITY BETWEEN THE LOWER OVAL CEILING AND THE CEILING ABOVE. REFER TO LUMINAIRE MOUNTING DETAIL FOR MORE INFORMATION.
9. PRESERVE AND PROTECT EXISTING 5-BUTTON CONTROL STATIONS. REPROGRAM EXISTING BUTTON STATIONS AS SHOWN IN THE LOW-VOLTAGE SWITCH SCHEDULE.
10. WALL MOUNT REMOTE POWER SUPPLY FOR TYPE T TRACK LIGHTING NEAR CAVEA DIMMING PANEL, COORDINATE EXACT LOCATION WITH OWNER PRIOR TO INSTALLATION.
11. SURFACE MOUNT REMOTE DRIVERS FOR TYPE WW WALL WASH LUMINAIRES ON TOP OF LOWER OVAL CEILING, COORDINATE EXACT LOCATION WITH OWNER PRIOR TO INSTALLATION.
12. EXISTING COVE LIGHTING SHALL REMAIN.
13. PROVIDE UL924 TRANSFER RELAY, BODINE GTD OR APPROVED EQUAL. RELAY SWITCHES LUMINAIRE TO EMERGENCY CIRCUIT UPON LOSS OF NORMAL POWER.
14. DISCONNECT AND REMOVE MULTI-HEAD ADJUSTABLE DOWNLIGHT SECTIONS BETWEEN LENSED LINEAR SECTIONS (KEYNOTE 3). PRESERVE AND PROTECT EXISTING BRANCH CIRCUIT FOR CONNECTION TO NEW LUMINAIRE.
15. RETURN EXISTING ETC TRACK, TRACKHEADS AND MONOPOINT LUMINAIRES TO OWNER.

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METRO STATE UNIVERSITY

CAVEA LIGHTING UPGRADE
PARTIAL LIGHTING PLAN
LEVEL 4

Date:	JULY 17, 2026
Scale:	1/4" = 1'-0"
Designed By:	SRJ
Drawn By:	MLB
Approved By:	SKB
Project No:	21607
Project Status:	100% CD

RMH-21607

SHT NO: E-114

REVISION

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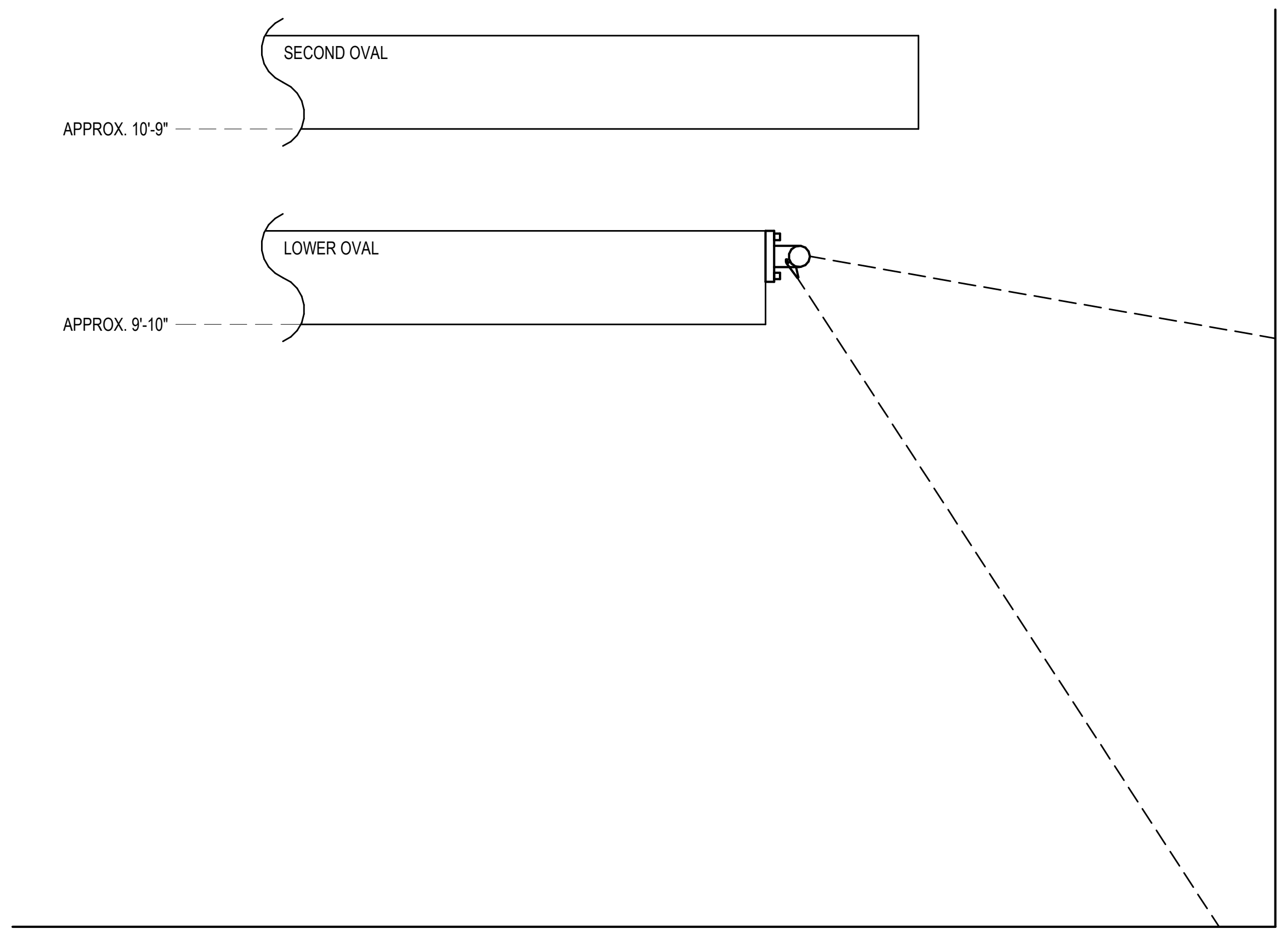
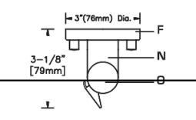


1
E-601
TYPE WW - WALL WASH DETAIL
SCALE: NONE



(E)CAVEA DIMMING PANEL

FED FROM: (E) L4B3
VOLTAGE: 208/120V, 3Ph, 5W
225A MLO
22,000 AIC

ETC 120V, 24 CIRCUIT DIMMER PANEL
DRd12 - 24 - 120 UNISON DIMMER PANEL CHASSIS
LOW-VOLTAGE DIMMING OPTION CARD
(NOTE 5)

DIMMER # & DMX ADDRESS	LOAD LUG # IN PANEL	ROOM NUMBER	ZONE	DESCRIPTION	DIMMER MODULE TYPE	LOAD TYPE	LOAD	PHASE	VIA EM TRANSFER SWITCH	NOTES
1	1	CAVEA	a	REAR WALL WASH	R20	0-10V	66	A		2
2	2	CAVEA	i	UPPER RING LTG		DMX	237			
3	7	CAVEA		SPARE	R20		0	A		4
4	8	CAVEA	d	CENTER DOWNLIGHTS		0-10V	112			
5	13	CAVEA		SPARE	R20		0	A		4
6	14	CAVEA	b	REAR DOWNLIGHTS		0-10V	136			
7	19	CAVEA		SPARE	D20		0	A		3
8	20	CAVEA		SPARE			0			
9	3	CAVEA		SPARE	R20		0	B		1
10	4	CAVEA	h	LOWER RING LTG		DMX	1068			
11	9	CAVEA	a	EXIT DOWNLIGHTS	R20	0-10V	17	B		4
12	10	CAVEA		SPARE			0			
13	15	CAVEA		SPARE	D20		0	B		1
14	16	CAVEA		SPARE			0			
15	21	CAVEA	e	NORTH TRACK LTG	R20	DMX	116	B		2
16	22	CAVEA	e	SOUTH TRACK LTG		DMX	116			
17	5	CAVEA	f	NORTH WALL WASH	R20	0-10V	180	C		4
18	6	CAVEA	g	SOUTH WALL WASH		0-10V	180			
19	11	CAVEA	c	FRONT DOWNLIGHTS	R20	0-10V	187	C		4
20	12	CAVEA		SPARE			0			
21	17	CAVEA		SPARE	D20		0	C		3
22	18	CAVEA		SPARE			0			
23	23	CAVEA		UNCONTROLLED PWR FOR SENSING FOR EM LTG	CC20	NON-DIM		C		1
24	24	CAVEA		UNCONTROLLED PWR FOR SENSING FOR EM LTG		NON-DIM				

CONTROLS KEY

CC	CONSTANT CURRENT - POWER ONLY - NO CONTROL
D20	FORWARD PHASE DIMMING
ELV10	REVERSE PHASE DIMMING
R20	RELAY

NOTES

1. EXISTING MODULE, EXISTING LOAD
2. EXISTING MODULE, NEW LOAD
3. EXISTING MODULE, LOAD REMOVED
4. NEW MODULE, NEW LOAD
5. LOAD REMOVED EXCEEDS LOAD ADDED, NO NET NEW LOAD

PANEL LOADING SUMMARY

LOAD TYPE	PH A	PH B	PH C	TOTAL	
LIGHTING	0.5	1.4	0.6	2.4	kVA
RECEPTACLE	0.0	0.0	0.0	0.0	kVA
COMPUTER	0.0	0.0	0.0	0.0	kVA
MOTOR	0.0	0.0	0.0	0.0	kVA
KITCHEN	0.0	0.0	0.0	0.0	kVA
HEAT	0.0	0.0	0.0	0.0	kVA
EQUIPMENT	0.0	0.0	0.0	0.0	kVA
OTHER	0.0	0.0	0.0	0.0	kVA
CONT LOAD	0.0	0.0	0.0	0.0	kVA
NONCOINCIDENT	0.0	0.0	0.0	0.0	kVA
PEAK LOAD	0.0	0.0	0.0	0.0	kVA
TOTAL	0.5	1.4	0.6	2.4	kVA

PHASE BALANCE	A-B	B-C	C-A	PF
(%)	-8	4	88	96

MIN PANEL AMPACITY	8	AMPERES

NEC DEMAND LOAD SUMMARY

LOAD TYPE	kW		POWER FACT		kVA		DEMAND FACTOR		CALCULATED LOAD	
LIGHTING	2.3	@	95%	=	2.4	@	125%	=	3.0	kVA
RECEPTACLES										
FIRST 10 kVA	0.0	@	95%	=	0.0	@	100%	=	0.0	kVA
REMAINDER	0.0	@	95%	=	0.0	@	50%	=	-	kVA
COMPUTER	0.0	@	95%	=	0.0	@	125%	=	0.0	kVA
MOTORS										
LARGEST	0.0	@	80%	=	0.0	@	125%	=	0.0	kVA
REMAINDER	0.0	@	80%	=	0.0	@	100%	=	0.0	kVA
KITCHEN	0.0	@	80%	=	0.0	@	100%	=	0.0	kVA
HEAT	0.0	@	100%	=	0.0	@	125%	=	0.0	kVA
EQUIPMENT	0.0	@	85%	=	0.0	@	100%	=	0.0	kVA
OTHER	0.0	@	85%	=	0.0	@	100%	=	0.0	kVA
CONT LOAD	0.0	@	85%	=	0.0	@	125%	=	0.0	kVA
NONCOINCIDENT	0.0	@	95%	=	0.0	@	0%	=	-	kVA
PEAK LOAD	0.0	@	90%	=	0.0	@	125%	=	0.0	kVA
0 % SPARE	0.0	@	90%	=	0.0	@	0%	=	0.0	kVA
TOTAL	2.3	kW			2.4	kVA			3.0	kVA

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Version 0525

LUMINAIRE SCHEDULE

Contact: Shara... Email: sjohnson@rmhgroup.com

	LED SOURCE				FINISH	MOUNTING	SPECIFICATION (NOTE 1)					
KEY	LUMENS	CCT	CRI	DESCRIPTION	(NOTE 2)	(NOTE 1)	MANUFACTURER	CATALOG NUMBER	DIMMING	VOLTS	VA	NOTES
D2	666	TUNABLE WHITE	80+	PINHOLE RECESSED RETROFIT DOWNLIGHT, MEDIUM 40 DEGREE DISTRIBUTION, DIFFUSE CLEAR FINISH EXISTING APPROXIMATE ROUGH-IN: 3.63" DIAMETER	WHITE	RECESSED	LUMENPULSE	LUMENCORE NANO DOWNLIGHT R - SB - RD - * - DWH - M - MWH - MWH - RM - 120 - DAMC	0-10V DOWN TO 1%	120	8.5	
RM	850	TUNABLE WHITE	80+	RECESSED LINEAR SLOT WITH MULTIHEAD ADJUSTABLE DOWNLIGHTS AT 9" SPACING; PROVIDE MODIFIED TRIM TO COVER EXISTING ROUGH-IN EXISTING APPROXIMATE ROUGH-IN: 5.43" OPENING x 3' LENGTH	BLACK	RECESSED	ZANIBONI	D4-D03RM - 13 - TD - 80 - 60 - S - BK - Z0 - ** - R0 - D	0-10V DOWN TO 1%	UNV	9W EACH DOWNLIGHT	
RW	2,661	TUNABLE WHITE	80+	SEMI-RECESSED RETROFIT WALLWASH WITH ASYMMETRIC WALL WASH DISTRUBUTION; PROVIDE MODIFIED TRIM TO COVER EXISTING ROUGH-IN EXISTING APPROXIMATE ROUGH-IN: 9-1/8" x 23-1/2"	WHITE	RECESSED	ELLIPTIPAR	S224 - S24 - T -22 - M - XX - 0 - TWH - DZ	0-10V DOWN TO 1%	UNV	22	
T	N/A	N/A	N/A	SURFACE MOUNT TRACK; PROVIDE COMPONENTS AND ACCESSORIES REQUIRED FOR A COMPLETE SYSTEM; PROVIDE POWER SUPPY FOR EACH TRACK CONFIGURATION	WHITE	SURFACE	WAC	VENTRIX SYSTEM TRACK: S2 - CPS** POWER SUPPLY: S2CPU - R - 320 - X - BK	--	120	--	
T1	1,745	TUNABLE WHITE	80+	TRACKHEAD WITH FIELD ADJUSTABLE OPTICS	WHITE	TRACK	WAC	SILO S2SA - 2 - F - TWA - WT	DMX DOWN TO 1%	120	29	
WW	1,903	TUNABLE WHITE	80+	SURFACE MOUNT WALL WASH WITH ASYMMETRIC WALL WASH DISTRIBUTION; REMOTE DRIVER	WHITE	SURFACE	ELLIPTIPAR	S100 - H-08 - E - 22 - M - 00 - 0 - TWH - DZ	0-10V DOWN TO 1%	UNV	15	

LUMINAIRE SCHEDULE NOTES:

1. COORDINATE VOLTAGE AND MOUNTING CONFIGURATION WITH PLANS.
2. CONFIRM FINISHES WITH OWNER.
3. CONFIRM EXACT EXISTING ROUGH-IN DIMENSIONS PRIOR TO PROVIDING LUMINAIRE SUBMITTALS AND SHOP DRAWINGS

LIGHTING CONTROL SEQUENCE OF OPERATION SCHEDULE

KEY	DESCRIPTION	CONTROL SYSTEM TYPE	VACANCY (MANUAL ON, AUTO OFF)	OCCUPANCY (AUTO ON, AUTO OFF)	TIMECLOCK	LOCAL MANUAL SWITCHING	LOCAL MANUAL DIMMING	SENSOR TIME DELAY	PHOTOSENSOR DIMMING	PHOTOSENSOR ON/OFF	NOTES
S12	VACANCY SENSING, LOCAL MANUAL DIMMING	STAND ALONE	X				X	15 MINUTE			LIGHTS SHALL BE MANUAL ON AND AUTOMATIC OFF WHEN THE SPACE IS VACANT. LOCAL DIMMING CONTROL.

LOW-VOLTAGE SWITCH SCHEDULE

LV SWITCH TAG	BUTTON	DESCRIPTION	ZONES CONTROLLED		DIM LEVELS	COLOR TEMPERATURE / COLOR	NOTES
(E)	1	ALL ON	a - f	ALL ZONES	100%	3500K	1
	2	STANDARD 1	a	EXIT DOWNLIGHTS	100%	3500K	
			b	REAR DOWNLIGHTS	100%	3500K	
			c	FRONT DOWNLIGHTS	100%	3500K	
			d	CENTER DOWNLIGHTS	100%	3500K	
			e	TRACK LIGHTS (each head individually addressed)	100%	3500K	
			f	NORTH WALL WASH	100%	3500K	
			g	SOUTH WALL WASH	100%	3500K	
			h	REAR WALL WASH	100%	3500K	
			i	LOWER RING LTG	100%	NOTE 2	
			j	UPPER RING LTG	100%	NOTE 2	
	3	STANDARD 2	a	EXIT DOWNLIGHTS	100%	3500K	
			b	REAR DOWNLIGHTS	100%	3500K	
			c	FRONT DOWNLIGHTS	100%	3500K	
			d	CENTER DOWNLIGHTS	100%	3500K	
			e	TRACK LIGHTS (each head individually addressed)	100%	3500K	
			f	NORTH WALL WASH	100%	3500K	
			g	SOUTH WALL WASH	0%	3500K	
			h	REAR WALL WASH	0%	3500K	
			i	LOWER RING LTG	100%	NOTE 2	
			j	UPPER RING LTG	100%	NOTE 2	
	4	SCREENING	a	EXIT DOWNLIGHTS	0%	3500K	
			b	REAR DOWNLIGHTS	10%	3500K	
			c	FRONT DOWNLIGHTS	10%	3500K	
			d	CENTER DOWNLIGHTS	10%	3500K	
			e	TRACK LIGHTS (each head individually addressed)	0%	3500K	
			f	NORTH WALL WASH	0%	3500K	
			g	SOUTH WALL WASH	0%	3500K	
			h	REAR WALL WASH	0%	3500K	
			i	LOWER RING LTG	100%	NOTE 2	
			j	UPPER RING LTG	100%	NOTE 2	
	5	ALL OFF	a - f	ALL ZONES	0%	3500K	

GENERAL NOTES:

1. COORDINATE FINAL SCENE DIM LEVELS AND COLOR TEMPERATURES WITH OWNER DURING PROGRAMMING.
2. COORDINATE COLOR SETTINGS WITH OWNER DURING PROGRAMMING.

REV	DATE	DESCRIPTION

SHARA RENEE JOHNSON
47675
07.17.2026

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METRO STATE
UNIVERSITY

CAVEA LIGHTING UPGRADE
ELECTRICAL SCHEDULES

Date:	JULY 17, 2026
Scale:	NONE
Designed By:	SRJ
Drawn By:	MLB
Approved By:	SKB
Project No:	21607
Project Status:	100% CD

RMH-21607

SHT NO:
E-601

REVISION

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COMcheck-Web™

Compliance Certificate

Project Information

Energy Code: 2024 IECC
Project Title: 21607 - MSU Cavea
Location: Denver, Colorado
Climate Zone: 5b
Project Type: Alteration
Project No: 122635
Project Sub Type: Initial Buildout
Project Mechanical Type: With Central HVAC/SWH/Plant
All Electric: false
Is Renewable: false
Has Battery: false
Has Charger: false
Has Heat Pump false

Construction Site: **Owner/Agent:** **Designer/Contractor:**

Notes:

Energy Credits

Description	Credit
Not applicable	

Renewable energy

Number of Floors: 1
Area of Three Largest Floors: 0
Proposed Capacity: 0
Required Capacity: 0
Proposed Off-site Energy: 0
Required Off-site Energy: 0
Exception: None

Renewables Pass

Building Area

Description	Floor Area
1-School/University (School/University) : Nonresidential	1324

Interior Lighting Compliance Certificate

Project Information

Energy Code: 2024 IECC
Project Title: 21607 - MSU Cavea
Location: Denver, Colorado
Climate Zone: 5b
Project Type: Alteration

Construction Site: **Owner/Agent:** **Designer/Contractor:**

Allowed Interior Lighting Power

A Area Category	B Floor Area (ft2)	C Allowed Watts / ft2	D Allowed Watts
1-Conference Room (Common Space Types:Conference/meeting/multipurpose room)	1324	0.88	1165
Total Allowed Watts:			**1165**

Proposed Interior Lighting Power

A Fixture ID : Description / Lamp / Wattage Per Lamp / Ballast	B # of Fixture	C Fixture Watt.	D (B X C)
1-Conference Room (Common Space Types:Conference/meeting/multipurpose room)			
D2: D2: LED:	40	8.5	340
RW: RW: LED:	3	22	66
RM: RM: LED:	12	6.2	74.4
WW: WW: LED:	12	15	180
C: C: LED:	168	3	504
T1: T1: LED:	12	29	Exempt
Exemption:Lighting for photographic processes.			
Total Proposed Watts:			**1164.4**

Proposed Interior Lighting Controls

Fixture	Lighting Control
1-Conference Room (Common Space Types:Conference/meeting/multipurpose room)	
D2: D2: LED:	Occupancy Sensor, Manual Control
RW: RW: LED:	Occupancy Sensor, Manual Control
RM: RM: LED:	Occupancy Sensor, Manual Control
WW: WW: LED:	Occupancy Sensor, Manual Control
C: C: LED:	Occupancy Sensor, Manual Control
T1: T1: LED:	Occupancy Sensor, Manual Control

Interior Lighting PASSES

Interior Lighting Compliance Statement

Compliance Statement: The proposed interior lighting design represented in this document is consistent with the building plans, specifications, and other calculations submitted with this permit application. The proposed interior lighting systems have been designed to meet the 2024 IECC requirements in COMcheckWeb and to comply with any applicable mandatory requirements listed in the Inspection Checklist.

Name - Title	Signature	Date
Mackenzie Brumage - Electrical Engineer	Mackenzie Brumage	09/21/2026

Inspection Checklist

Energy Code: 2024 IECC

Requirements: 61% were addressed directly in the COM*check* software

Text in the "Comments/Assumptions" column is provided by the user in the COMcheck Requirements screen. For each requirement, the user certifies that a code requirement will be met and how that is documented, or that an exception is being claimed. Where compliance is itemized in a separate table, a reference to that table is provided.

Plan Review

Section # & Req.ID	Plan Review	Complies?	Comments/Assumptions
C105.2 [PR3][1]	Plans, specifications, and/or calculations provide all information with which compliance can be determined for the interior lighting and electrical systems and equipment and document where exceptions to the standard are claimed. Information provided should include interior lighting power calculations, wattage of bulbs and ballasts, transformers and control devices.	☐ Complies ☐ Does Not Comply ☐ Not Observable ☐ Not Applicable	**Location on plans/spec:** Specs

Additional Comments/Assumptions:

1	High Impact (Tier 1)	2	Medium Impact (Tier 2)	3	Low Impact (Tier 3)

Rough-In Electrical Inspection

Section # & Req.ID	Rough-In Electrical Inspection	Complies?	Comments/Assumptions
C405.2.1, C405.2.1.1 [EL7][1]	Occupancy sensors installed in classrooms/lecture/training rooms, computer rooms/data center, conference/meeting/multipurpose rooms, copy/print rooms, lounges/breakrooms, medical supply rooms, enclosed offices, laundry areas, open plan office areas, restrooms, storage rooms, telemedicine rooms, locker rooms, corridors, warehouse storage areas, and other spaces <= 300 sqft that are enclosed by floor-to-ceiling height partitions.	☐ Complies ☐ Does Not Comply ☐ Not Observable ☐ Not Applicable	**Exception:** Requirement does not apply.
C405.7 [EL3][2]	Low-voltage dry-type distribution electric transformers meet the minimum efficiency requirements of Table C405.6.	☐ Complies ☐ Does Not Comply ☐ Not Observable ☐ Not Applicable	
C405.8 [EL4][2]	Electric motors meet the minimum efficiency requirements of Tables C405.7(1) through C405.7(4). Efficiency verified through certification under an approved certification program or the equipment efficiency ratings shall be provided by motor manufacturer (where certification programs do not exist).	☐ Complies ☐ Does Not Comply ☐ Not Observable ☐ Not Applicable	
C405.10 [EL1][2]	Total voltage drop across the combination of feeders and branch circuits <= 5%.	☐ Complies ☐ Does Not Comply ☐ Not Observable ☐ Not Applicable	
C405.10.1, C405.10.2 [EL5][2]	Escalators and moving walks comply with ASME A17.1/CSA B44 and have automatic controls configured to reduce speed to the minimum permitted speed in accordance with ASME A17.1/CSA B44 or applicable local code when not conveying passengers.	☐ Complies ☐ Does Not Comply ☐ Not Observable ☐ Not Applicable	
C405.12, C405.12.1 [EL6][2]	50% of 15/20 amp receptacles installed in enclosed offices, conference rooms, copy rooms, break rooms, classrooms and workstations and > 25% of branch circuit feeders for modular furniture will have automatic receptacle control in accordance with C405.12.1.	☐ Complies ☐ Does Not Comply ☐ Not Observable ☐ Not Applicable	
C405.2.1.2 [EL8][1]	Occupancy sensors control function in warehouses: In warehouses, the lighting in aisleways and open areas is controlled with occupant sensors that automatically reduce lighting power by 50% or more within 20 minutes of when the areas are unoccupied. The occupant sensors control lighting in each aisleway independently and do not control lighting beyond the aisleway being controlled by the sensor. Lights not turned off by occupant sensors is done so by time-switch.	☐ Complies ☐ Does Not Comply ☐ Not Observable ☐ Not Applicable	**Exception:** Requirement does not apply.
C405.2.1.3 [EL9][1]	Occupant sensor control function in open plan office areas: Occupant sensor controls in open office spaces >= 300 sq.ft. have controls 1) configured so that general lighting can be controlled separately in control zones with floor areas <= 600 sq.ft. within the space, 2) general lighting in each zone permitted to turn on upon occupancy in control zone, 3) automatically turn off general lighting in all control zones within 20 minutes after all occupants have left the space, 4) are configured so that general lighting power in each control zone is reduced by >= 80% of the full zone general lighting power within 20 minutes of all occupants leaving that control zone.	☐ Complies ☐ Does Not Comply ☐ Not Observable ☐ Not Applicable	**Exception:** Requirement does not apply.
C405.2.10, C405.2.10.1, C405.2.10.2 [EL10][1]	Sleeping or dwelling units provided with lighting controls	☐ Complies ☐ Does Not Comply ☐ Not Observable ☐ Not Applicable	**Exception:** Requirement does not apply.

Section # & Req.ID	Rough-In Electrical Inspection	Complies?	Comments/Assumptions
C405.2.2, C405.2.2.1 [EL11][2]	Each area not served by occupancy sensors (per C405.2.1.1) have time-switch controls and functions detailed in sections C405.2.2.1.	☐ Complies ☐ Does Not Comply ☐ Not Observable ☐ Not Applicable	**Exception:** Requirement does not apply.
C405.2.3.1 [EL12][1]	Spaces required to have dimming controls have a manual control that allows the occupant to reduce the connected lighting load to 10% of full power.	☐ Complies ☐ Does Not Comply ☐ Not Observable ☐ Not Applicable	**Location on plans/spec:** E-114
C405.2.4, C405.2.4.1, C405.2.4.2 [EL13][2]	Daylight zones provided with individual controls that control the lights independent of general area lighting. See code section C405.2.3 Daylight-responsive controls for applicable spaces, C405.2.3.1 Daylight responsive control function and section C405.2.3.2 Sidelit zone.	☐ Complies ☐ Does Not Comply ☐ Not Observable ☐ Not Applicable	**Exception:** Requirement does not apply.
C405.2.5 [EL14][1]	Additional interior lighting power allowed for special functions per the approved lighting plans and is automatically controlled and separated from general lighting.	☐ Complies ☐ Does Not Comply ☐ Not Observable ☐ Not Applicable	**Location on plans/spec:** N/A

Additional Comments/Assumptions:

1	High Impact (Tier 1)	2	Medium Impact (Tier 2)	3	Low Impact (Tier 3)

Final Inspection

Section # & Req.ID	Final Inspection	Complies?	Comments/Assumptions
C408.1.1 [FI58][1]	Building operations and maintenance documents will be provided to the owner. Documents will cover manufacturers' information, specifications, programming procedures and means of illustrating to owner how building, equipment and systems are intended to be installed, maintained, and operated.	☐ Complies ☐ Does Not Comply ☐ Not Observable ☐ Not Applicable	
C408.2.5 [FI76][3]	Furnished as-built drawings for electric power systems within 90 days of system acceptance.	☐ Complies ☐ Does Not Comply ☐ Not Observable ☐ Not Applicable	
C408.3 [FI4][1]	Lighting systems and receptacle controls have been tested to ensure proper calibration, adjustment, programming, and operation in accordance with C408.3.1.1 through C408.3.1.6.	☐ Complies ☐ Does Not Comply ☐ Not Observable ☐ Not Applicable	**Location on plans/spec:** Specs
C408.3.2.2 [FI77][3]	Furnished O&M instructions for systems and equipment to the building owner or designated representative.	☐ Complies ☐ Does Not Comply ☐ Not Observable ☐ Not Applicable	

Additional Comments/Assumptions:

1	High Impact (Tier 1)	2	Medium Impact (Tier 2)	3	Low Impact (Tier 3)

REV	DATE	DESCRIPTION
1	09/21/26	RESPONSE TO COMMENTS

SHARA RENEE JOHNSON
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09.21.2026

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METRO STATE
UNIVERSITY

CAVEA LIGHTING UPGRADE
LIGHTING COMCHECK

Date: JULY 17, 2026
Scale:
Designed By: SRJ
Drawn By: MLB
Approved By: SKB
Project No: 21607
Project Status: 100% CD
RMH-21607

SHT NO: E-602
REVISION 1

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